

The challenges of Antimicrobial Resistance (AMR) in hospital-acquired infections and the use of AI to overcome them

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Academic background

- **Rey Juan Carlos University (URJC), Madrid, Spain**

- B.Sc. in Telecommunications
- B.Sc. in Computer Science



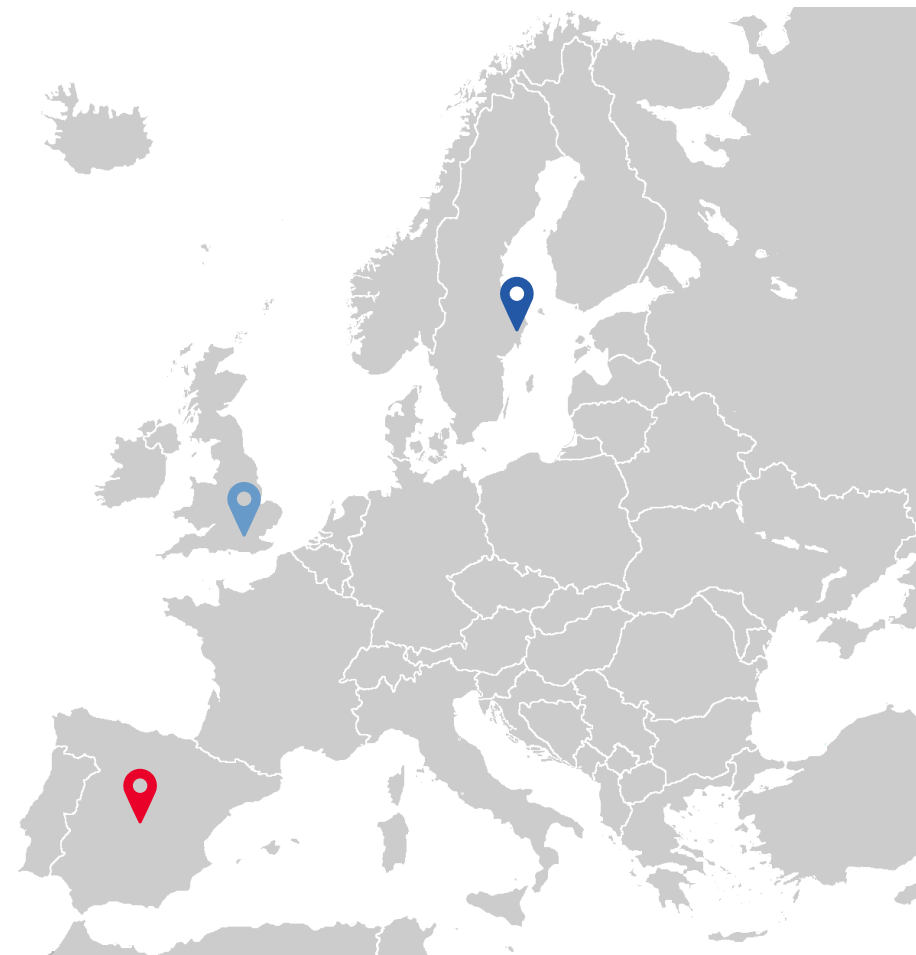
- **Royal Institute of Technology (KTH), Stockholm, Sweden**

- M.Sc. in Machine Learning



- **Imperial College London (ICL), London, United Kingdom**

- Ph.D. in Computer Science and Healthcare, EEE
- Research Associate in AI/ML,
- Research Fellow in AI/ML



Centre for Antimicrobial Optimisation

Innovation in technology and data usage

To improve the use of antimicrobials so treatment can be better individualised, patient care improved, and drivers of AMR reduced.

- **Harnessing data** at multiple levels.
- Exploring the **potential of Artificial Intelligence** (AI) for clinical decision support (CDSS)
- Diagnosis, prognosis, therapy selection and patient similarity retrieval.
- Optimization therapy **de-escalation** and cessation.
- Targeted, **precision dosing** tailored for the individual, the infection, the site, the organism, and the resistance profile
- Building **patient contact networks** to trace pathogen transmission in hospitals, identify emerging resistance patterns, and outbreaks.



The banner features a dark blue header with the text 'Centre for Antimicrobial Optimisation' and navigation links: 'About Us', 'Research', 'People', 'News', and 'Contact Us'. Below the header is a collage of three images: a petri dish with red agar and bacterial colonies, a person in a lab coat using a pipette on a microfluidic device, and a healthcare worker in a clinical setting wearing a mask and gloves. A teal bar at the bottom of the collage contains the text 'Optimising Antimicrobial Use To Address The Global Challenge of Antimicrobial Resistance'. Below the collage are three logos: 'Imperial College London', 'camo centre for antimicrobial optimisation', and the 'Department of Health & Social Care' logo. At the bottom of the banner are three small images with labels: 'Innovative Diagnostics' (a green microfluidic chip), 'Therapeutic Solutions' (a grid of yellow microfluidic chips), and 'Data Sciences' (a blue microfluidic chip).

Centre for Antimicrobial Optimisation

About Us | Research | People | News | Contact Us

Optimising Antimicrobial Use To Address The Global Challenge of Antimicrobial Resistance

Imperial College London

camo
centre for antimicrobial optimisation

Department of Health & Social Care

Innovative Diagnostics

Therapeutic Solutions

Data Sciences

CAMO-Net

The **Centres for Antimicrobial Optimization Network** is a welcome-funded, multidisciplinary global research partnership dedicated to addressing the global impact of Antimicrobial Resistance (AMR)

Mission

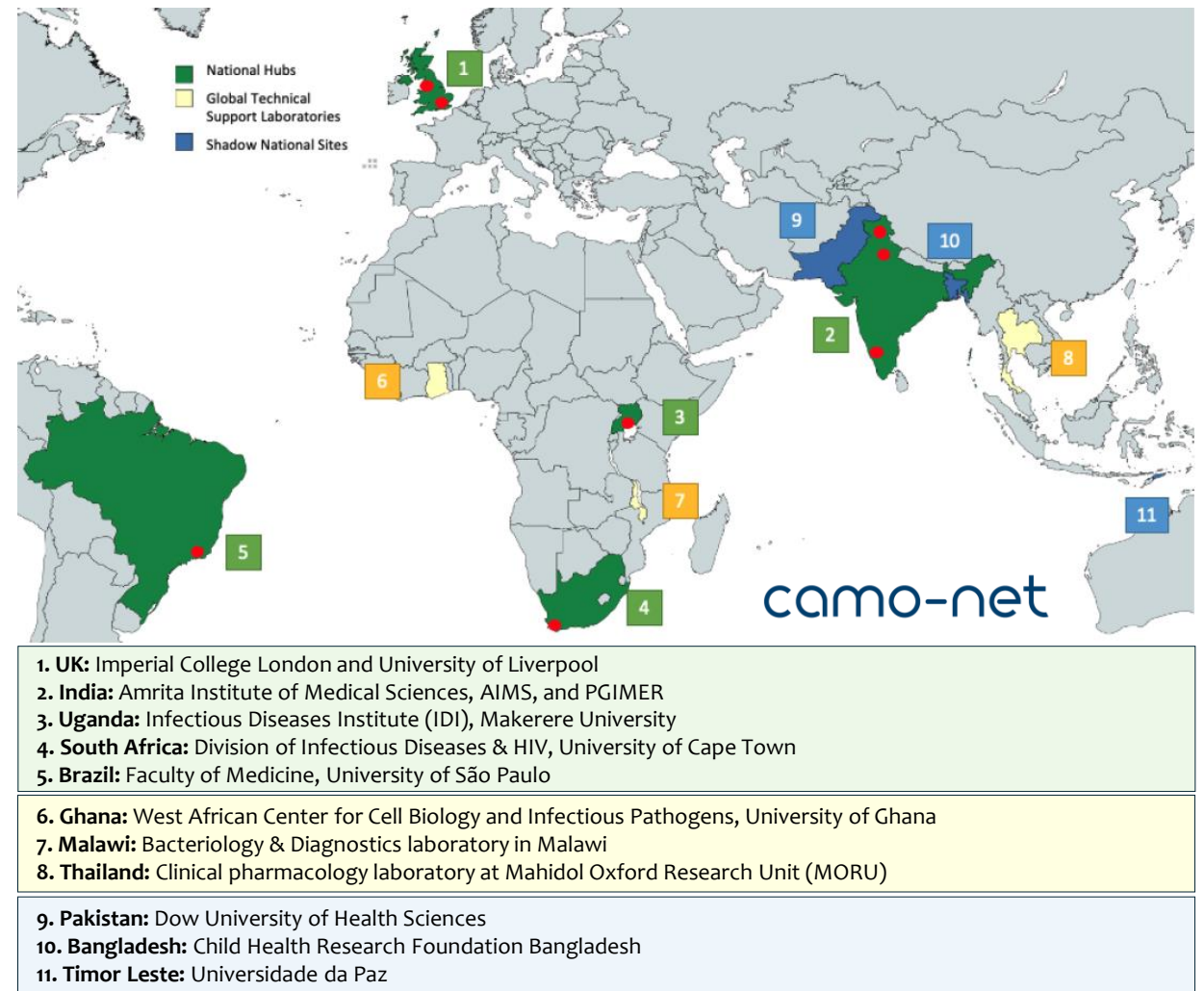
To create a sustainable and equitable research ecosystem that optimizes antimicrobial use in humans, developing context-relevant tools, technologies, and practices.

Core Focus Areas

- **Technology & innovation** for optimized prescribing
- Understanding context, culture, and behaviour
- Improving medicines management

Global Structure

A collaborative network of 5 National Hubs (UK, Brazil, India, South Africa, Uganda), 3 Technical Support Centres, and 3 Shadow National Sites.



Contents

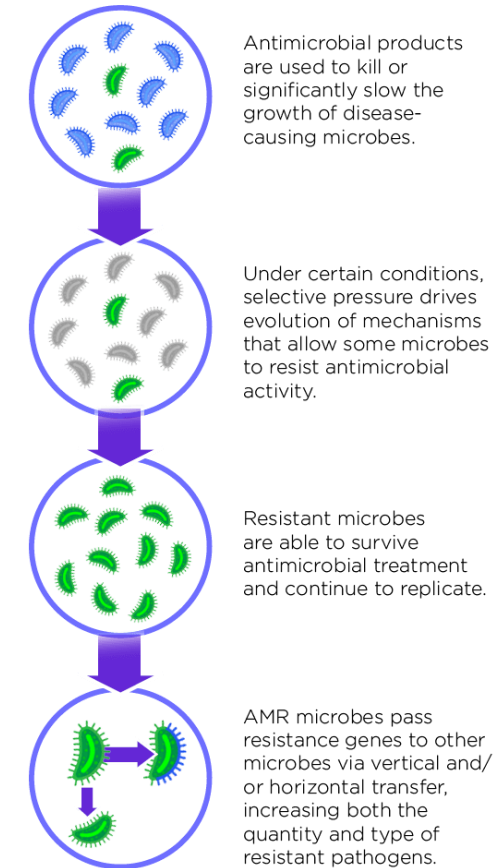
- **Introduction**
 - AMR landscape
 - Leading AMR drivers
 - The infection management pathway
- **Enhancing clinical decision-making with AI and ML**
 - Introduction to Clinical Decision Support Systems
 - Supervised machine learning for diagnosis/prognosis
 - Leveraging clinical and biochemistry data
 - Unlocking the power of longitudinal data analysis
 - Innovative approaches for label-free, multi-output, and reusable models
 - Limitations with supervised machine learning
 - The case-based reasoning methodology
 - Proof of concept: A real-life example with Dengue
- **Conclusions**

What is AMR?

Definition

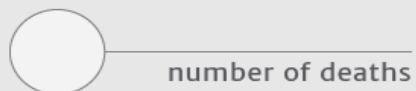
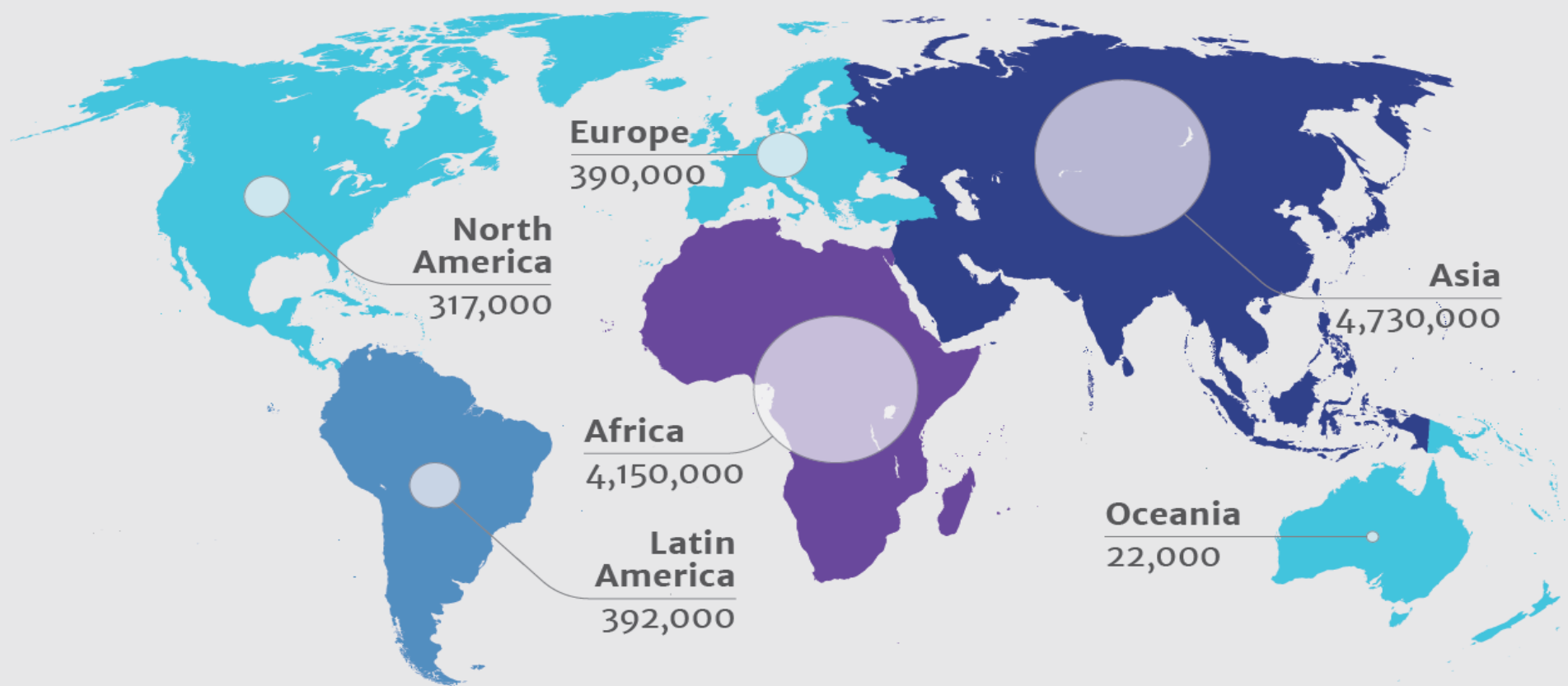
Antimicrobial resistance (AMR) occurs when **microorganisms**, such as bacteria, viruses, fungi, and parasites, **evolve to resist the medicines**, including antibiotics, antivirals, and antifungals, that were once used to treat them.

- Infections become harder to treat
- Common infections can become life-threatening (e.g. UTIs)
- Routine medical procedures become riskier (e.g. cancer)
- Severe impact on vulnerable populations (e.g. immunocompromised)
- Leads to a greater risk of disease spread (e.g. outbreaks)
- Increased healthcare costs
- Increased morbidity and mortality

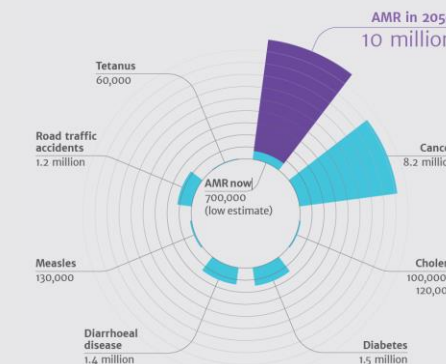
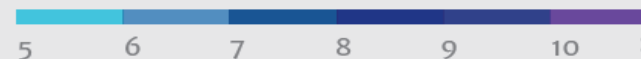


Deaths attributable to AMR every year by 2050

O'Neill, J. (2016). *Tackling Drug-Resistant Infections Globally: Final Report and Recommendations*. The Review on Antimicrobial Resistance. London: HM Government and Wellcome Trust.

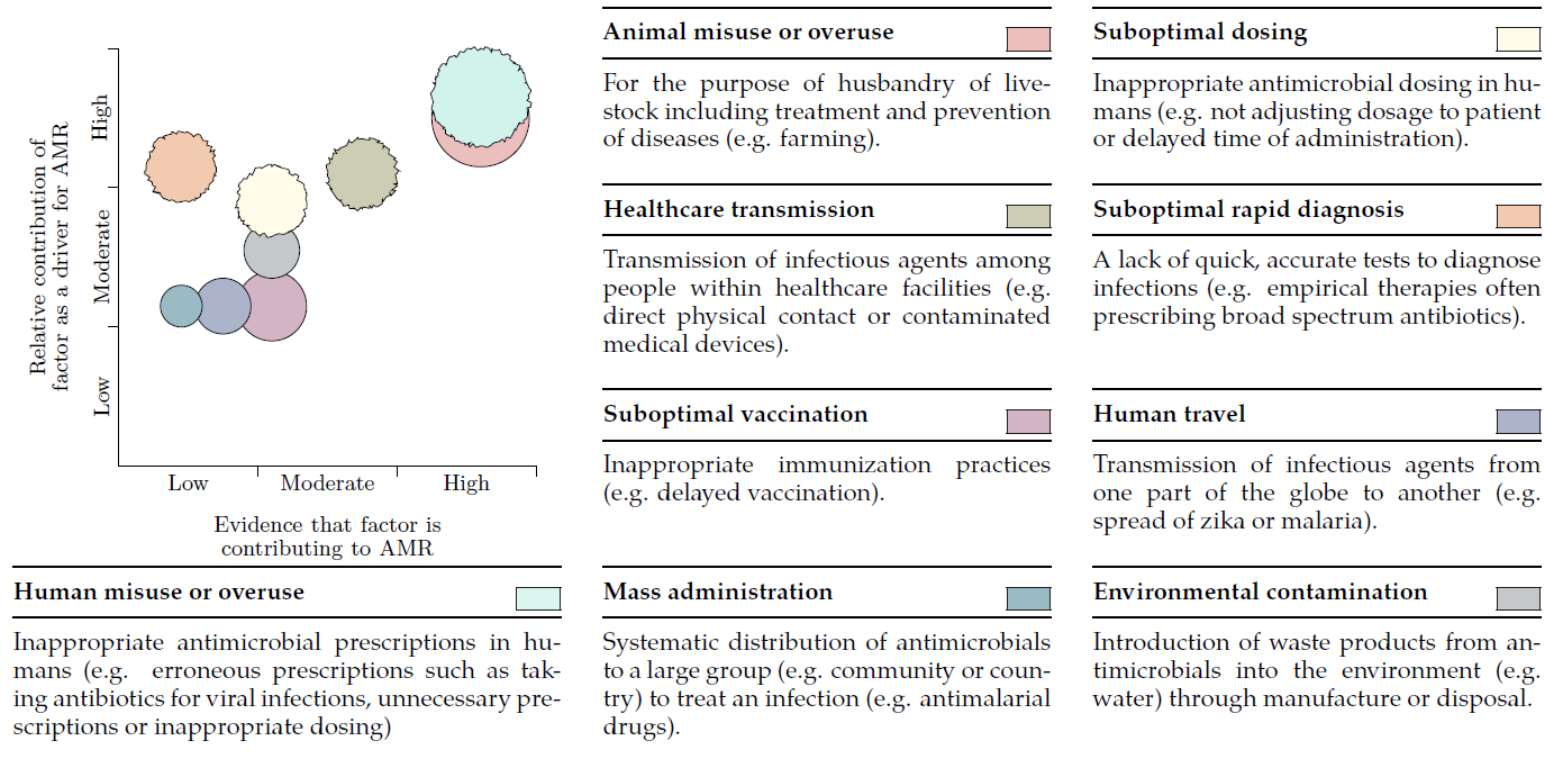


Mortality per 10,000 population



Leading drivers or AMR

Table 2.1: Leading drivers of antimicrobial resistance.



An info-graphic to show the considered potential contribution of each factor as a driver for antimicrobial resistance including further explanation and examples. The diameter of bubble represents the potential population affected [23, 25]. Irregular circles represent drivers that could be addressed through the implementation of decision support systems in healthcare settings.

What is a CDSS?

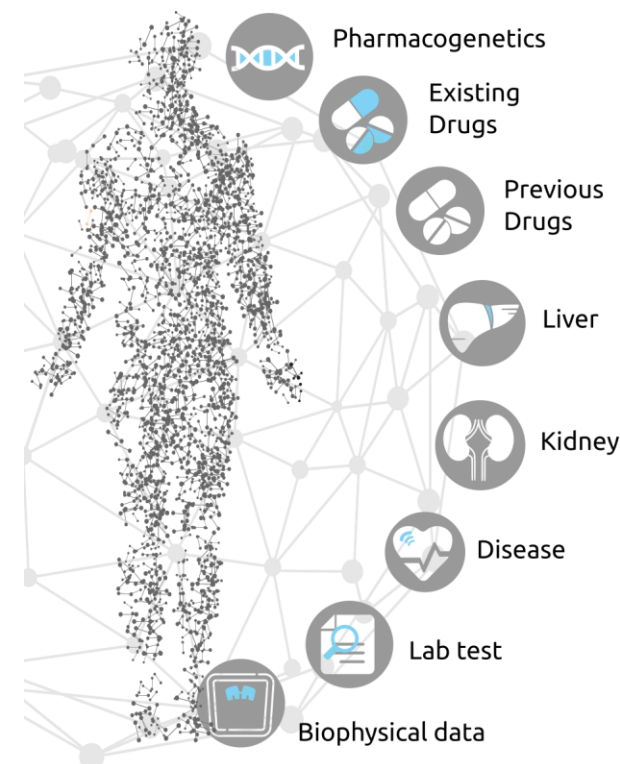
Definition

“A clinical decision support system (CDSS) is **a health information technology** that provides clinicians, staff, patients, and other individuals with **knowledge** and person-specific information to help health and health care. CDSS encompasses a variety of tools to **enhance decision-making in the clinical workflow.**”

– Wikipedia

Drug discovery	Data access / Alerts / Reminders
Plan clinical trials	Diagnostic / Prognosis support
Telehealth	Classification / Risk stratification
Genomic decision support	Therapy recommendation
Infection control	Precision dosing
Surveillance	Imaging and text

Let's focus on those with direct application in **daily clinical practice.**



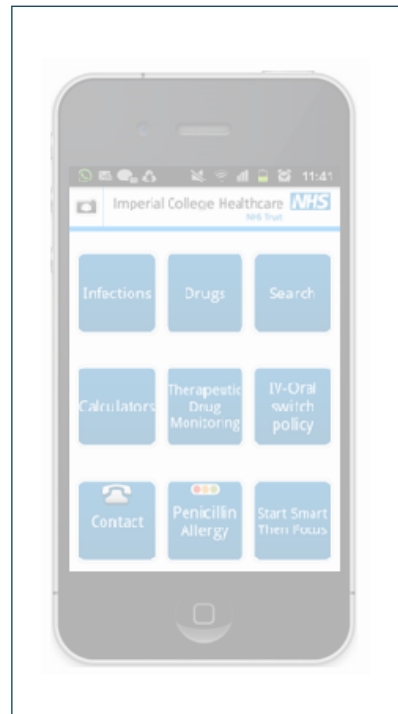
The evolving landscape of CDSSs

A **paper** pocket guide

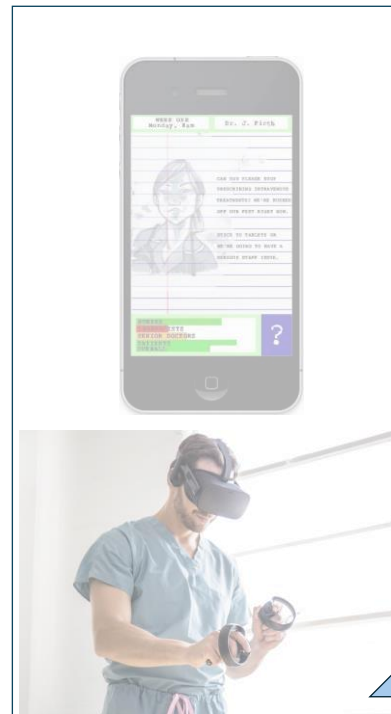


A **digital** guide

A **smartphone** app

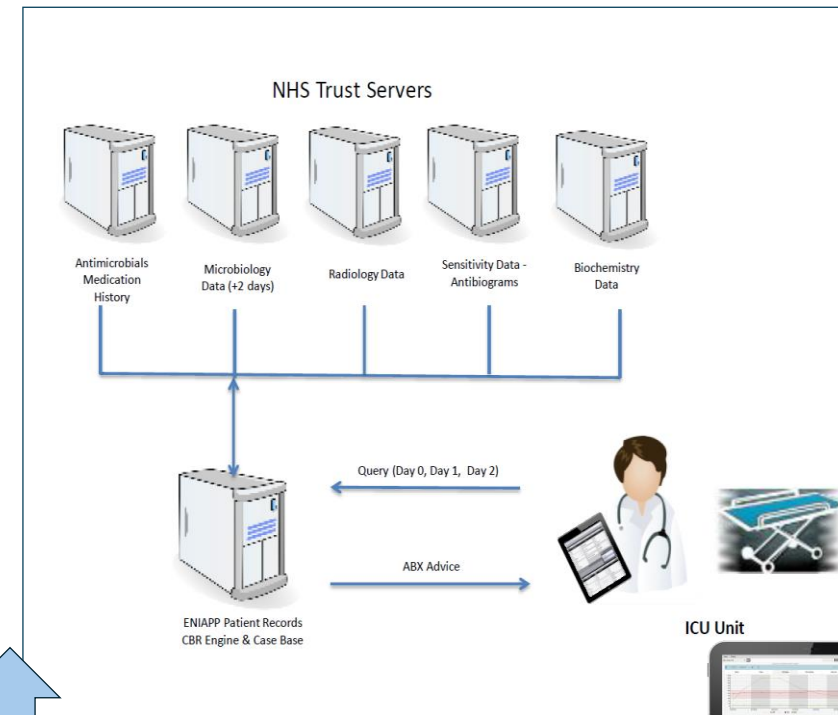


Incentives and **gaming**



A mobile **real-time** EHR data-driven CDSS for clinical decision support.

1, 2, 3



Surge in
EHR



epicImpoc
enhanced, personalized and integrated care
for infection management at point of care

1. T. Rawson et al – Supervised machine learning for the prediction of infection on admission to hospital: a prospective observational cohort study – JAC (2018)
2. B. Hernandez et al – Data-driven Web-based Intelligent Decision Support System for Infection Management at Point-Of-Care: CBR Benefits and Limitations – HINF (2017)
3. B. Hernandez et al – Supervised Learning for Infection Risk Inference Using Pathology Data – BMC Medical Informatics and Decision Making (2017)

Enhancing healthcare with CDSSs



Management to facilitate access to records, laboratory results, and communication.



Facilitate **data collection** to enhance decisions, send **reminders** or clinical **alerts**



Patient engagement to improve understanding and shape future patient behaviour



Assess or predict complications using machine learning methods on tabular patient data



Review of **past similar cases** to improve decisions matching clinicians' reasoning process



AMR surveillance to strengthen knowledge for action in support of strategies to improve guidelines



Support **medication** from allergy checking, and therapy selection to adjusting antimicrobial doses



Using **NLP** to extract medical information from free-text clinical notes



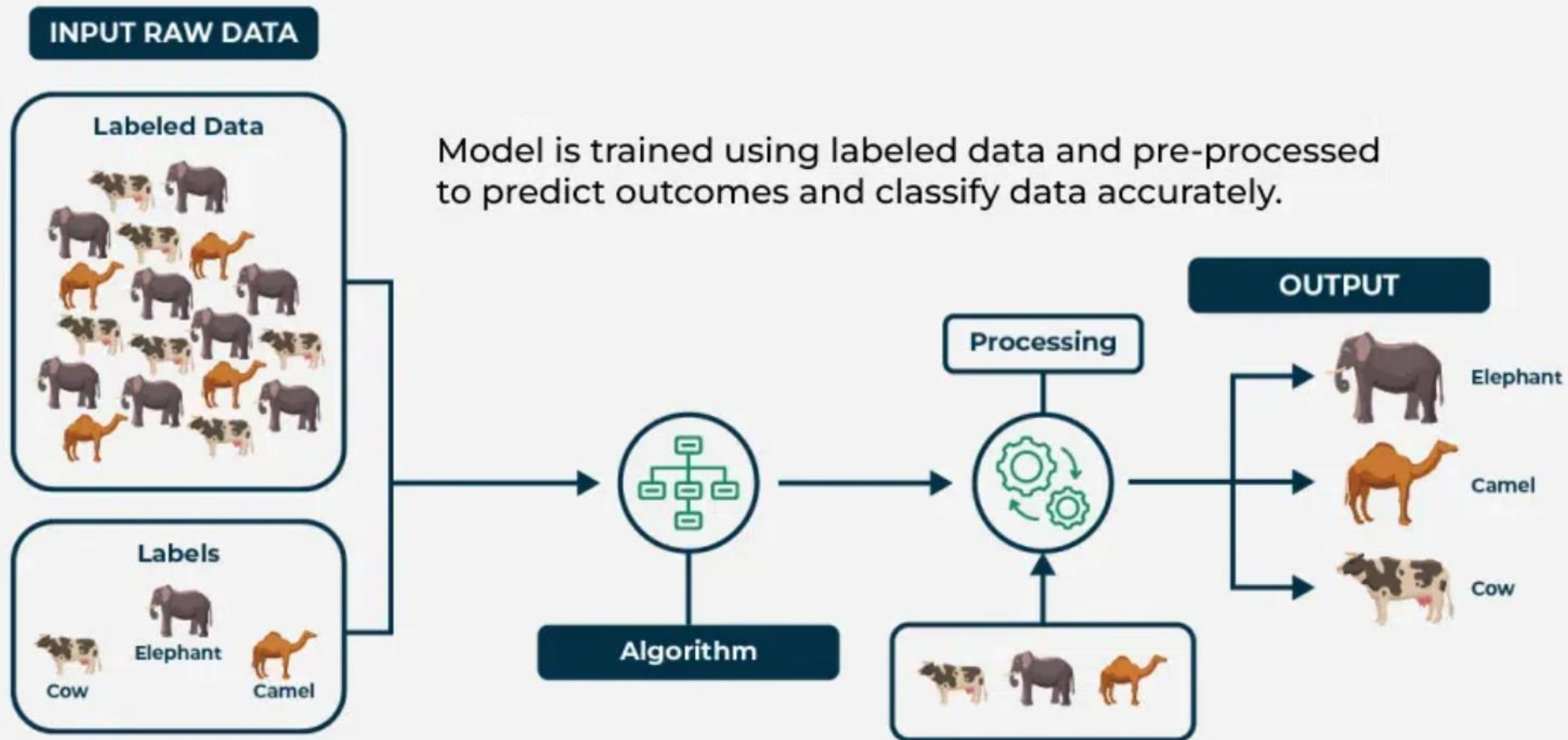
Image processing to extract useful information from medical imaging (X-ray, CT, MRI)



+ more!



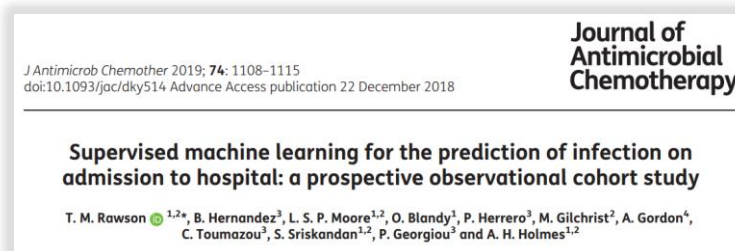
Supervised learning



Leveraging AI/ML for diagnosis and prognosis



- Diagnosis of **infection** (and site)
 - Assess positive microbiology
 - Using only 6 routinely available biomarkers
 - Handling a largely imbalanced dataset
 - Robust if at least four biomarkers available
 - Validated prospectively



Bernard Hernandez

Research Fellow
Department of Infectious Diseases
Imperial College London

Infection

>500.000 daily profiles
2.7% prevalence

6 biomarkers
CRP, WBC, BIL, CRE, ALT, ALP

SVM
AUC ROC 0.85 (95% CI: 0.84 - 0.86)
SENS 0.75 | SPEC 0.91

104 patients
35% prevalence at 72h
42% had microbiological tests

AUC ROC 0.84 (95% CI: 0.76 - 0.91)
SENS 0.89 | SPEC 0.63 at 0.81

1. B. Hernandez et al – *Data-driven web based intelligent CDSS for infection management at the point of care* – PhD Thesis Imperial College London (2019)
2. T. Rawson et al – *Supervised machine learning for the prediction of infection on admission to hospital: a prospective observational cohort study* – JAC (2018)
3. B. Hernandez et al – *Supervised learning for infection risk inference using pathology data* – BMC Medical Informatics and Decision Making (2017)

Leveraging AI/ML for diagnosis and prognosis



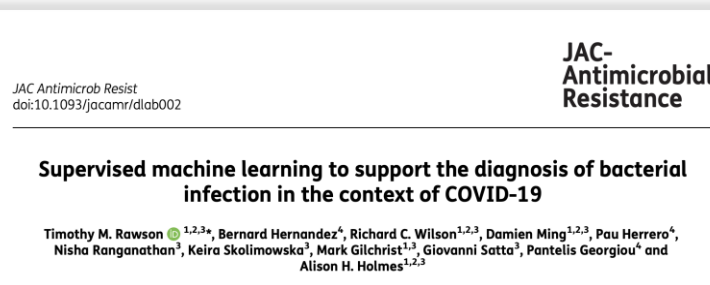
- Diagnosis of **infection** (and site)
- Quickly **repurposed** for **COVID-19**!
 - Assess positive microbiology
 - Using a single blood panel (Full Blood Count)
 - Validated prospectively



Bernard Hernandez

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SARS-CoV-2

1186 patients
65% prevalence
14% had microbiological tests

21 biomarkers
Full Blood Count (FBC)

SVM
AUC ROC 0.91 (95% CI: 0.76 - 0.91)
SENS 0.80 | SPEC 0.89

54 patients
52% prevalence

AUC ROC (0.96% CI: 0.90 - 1.00)
SENS 0.75 | SPEC 0.90

Unlocking temporal data analysis



- Models **accounting for temporal patterns** provide better results than standard static approaches.
- Performed better than **static approaches**
AUC ROC 0.74 (95% CI: 0.70-0.78) ↑
- Decrease in performance after **removing temporal** information (randomly shuffling timepoints)
AUCROC 0.86 (95% CI: 0.85-0.89) ↓
- Decrease in performance in **hospital-acquired BSIs**
AUC ROC 0.65 (95% CI: 0.80-0.70) ↓
- The **most important features** were the **dynamics** of C-Reactive protein (CRP), neutrophils and platelets

Bloodstream Infection

20,850 patients
18% prevalence

31 biomarkers
FBC, U&E, CRP, Lactate and COVID-19

LSTM
AUC ROC 0.97 (95% CI: 0.96 – 0.97)
SENS 0.94 | SPEC 0.93
PPV 0.58 | NPV 0.99

+ time-aware split
+ consistent (genres, age, specialties)
+ using temporal dynamics



Damien K. Ming

Clinical Research Fellow
Department of Infectious Diseases
Imperial College London

D Ming et al – *Utilising routinely collected clinical data through time series deep learning to improve identification of bacterial bloodstream infections and management of blood culture results* – Lancet Digital Health (2025)

Limitations of supervised learning approaches



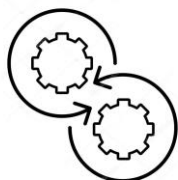
Needs the **ground truth** or label for which the model is optimised.



Needs to be **(re)trained** for each label or **phenotype**.



It returns a probability which might be difficult to **interpretate**.



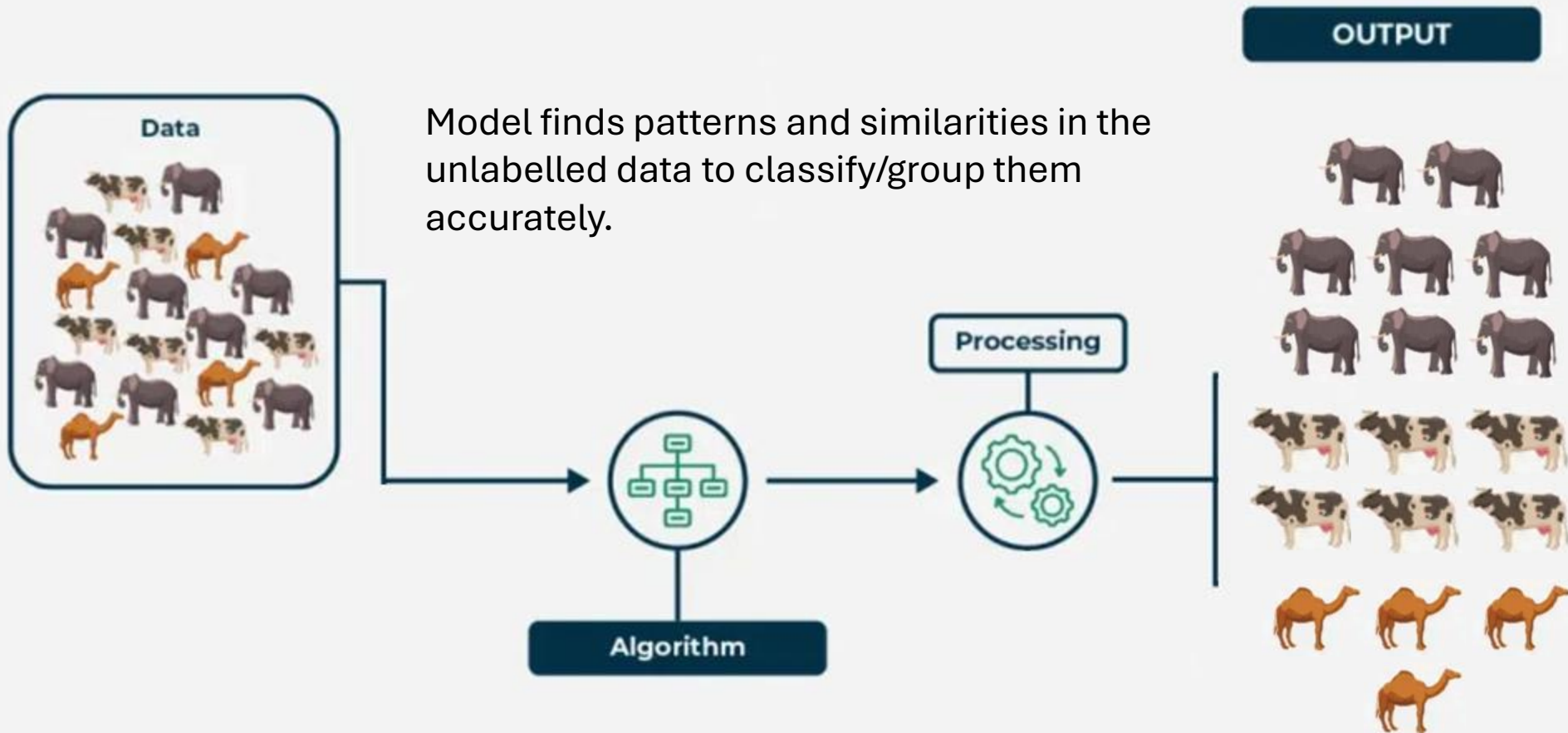
Outputs from **different algorithms** (e.g. shock and fluid accumulation) **must be consistent**.



Lacks **explainability**; that is, what are the underlying reasons motivating such probabilities.

Could we **identify similarity patterns** between patients **without** using predefined **labels**?

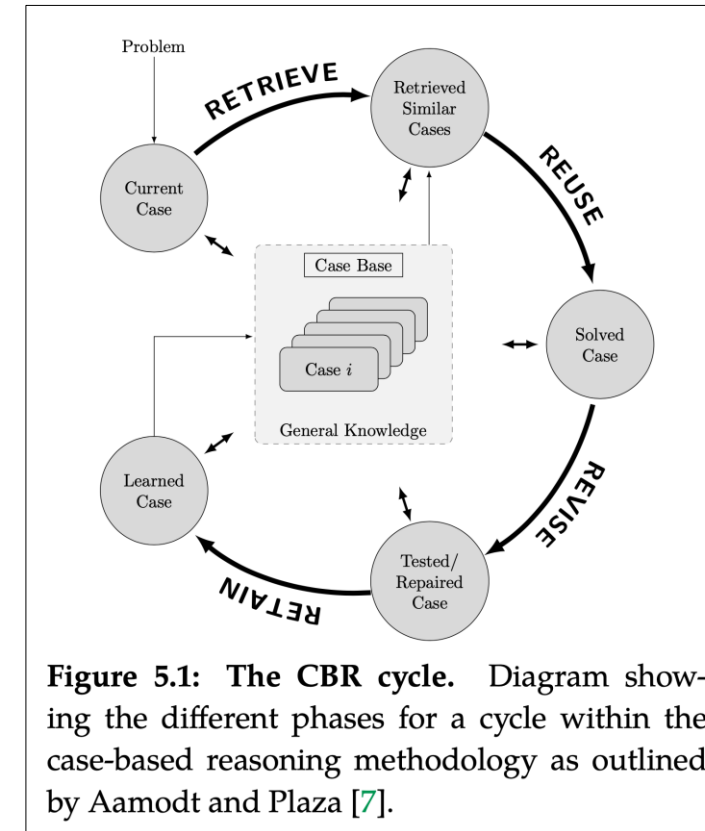
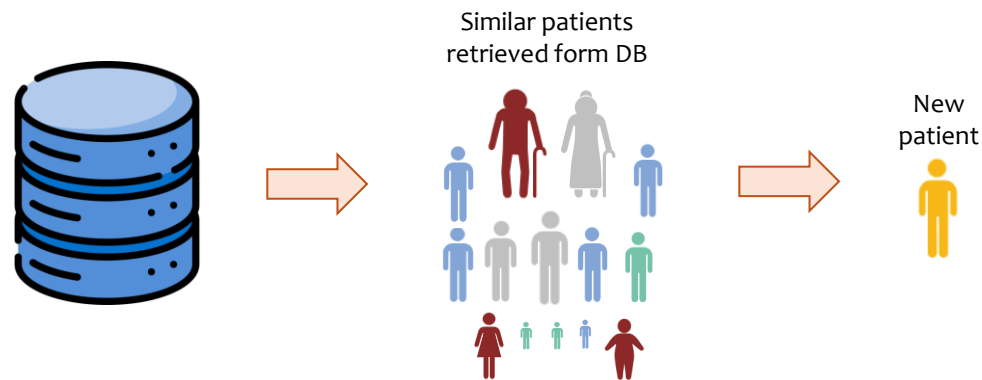
Unsupervised learning



What is Case-Based Reasoning?

Definition

Case-Based Reasoning (CBR) is a problem-solving and decision-making **methodology** that relies on using **past experiences** (cases) to solve new problems or **make decisions** in similar situations. It aligns with the cognitive process and **reasoning of clinicians** who make decisions based on knowledge acquired from their experience.





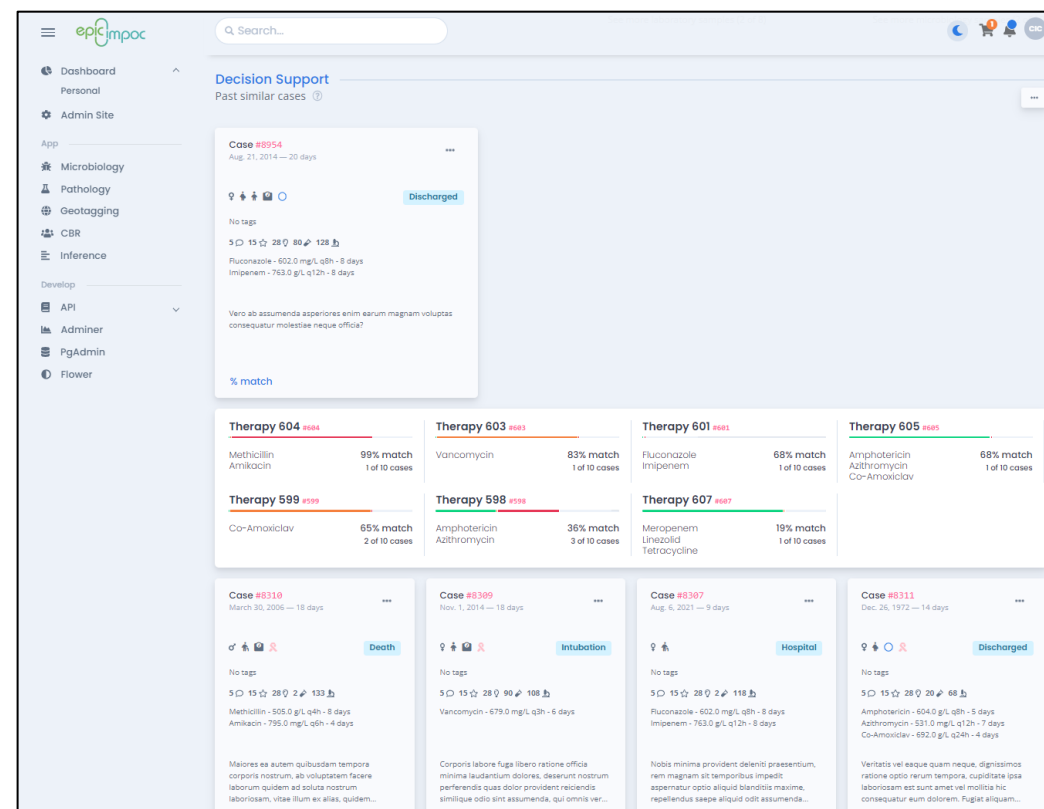
Determining the most appropriate therapy

- Using **Case-Based Reasoning** (CBR)
- Clinical setting
 - Presentation with possible infection
 - Physician entry and trust servers' data
- **Intensive Care:** CBR vs expert option
 - High level of agreement (90%)
- **Non-critical care:** CBR vs susceptibility data
 - Reduced antimicrobial spectrum
Reduced Antimicrobial Spectrum Index (from 8 to 6)
More likely to be classified as Access class (49% vs 35%)



Timothy M. Rawson

Honorary Clinical Lecturer
Department of Infectious Diseases
Imperial College London

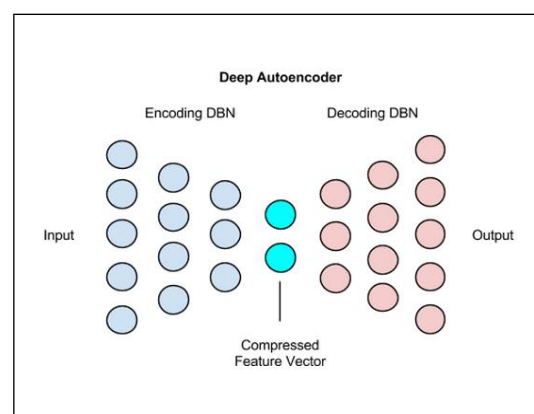
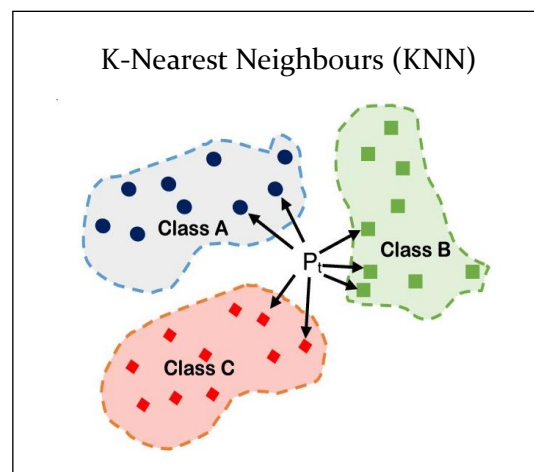


1. T. Rawson et al – A real-world evaluation of a CBR to support antimicrobial prescribing decisions in acute care – CID (2020)
2. B. Hernandez et al – Data-driven web-based intelligent CDSS: CBR benefits and limitations – Health informatics (2017)

Enhancing interpretability and generalizability



- Create **2D embeddings** where similar patients are close to each other.
- Retrieve K **similar patients** and **visualise** vital signs, symptoms or classifications estimates.
- Increase **generalizability** by allowing multimodal data including sequential (timeseries) pattern extraction



Bernard Hernandez

Research Fellow

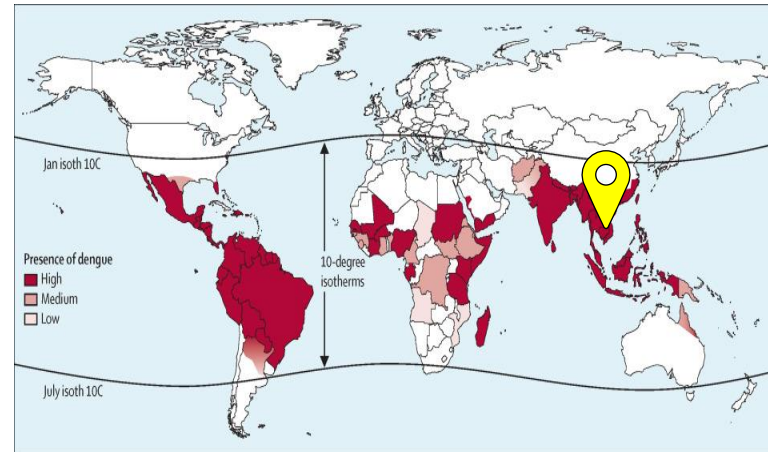
Department of Infectious Diseases

Imperial College London

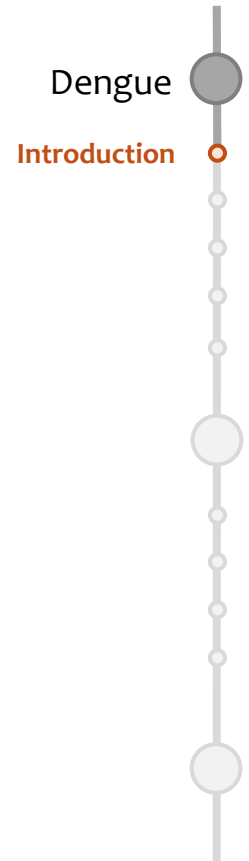
Proof of concept: Dengue

Introduction

- Dengue fever is a **mosquito-borne** viral infection.
- It is endemic in over 100 countries, with higher prevalence in **tropical** and **subtropical** regions.
- Approximately 51 million symptomatic cases each year, with **seasonal epidemics** and high caseloads imposing a huge strain on local healthcare services.
- Major public health concern with **increasing incidence** (research suggesting climate change influences transmission).



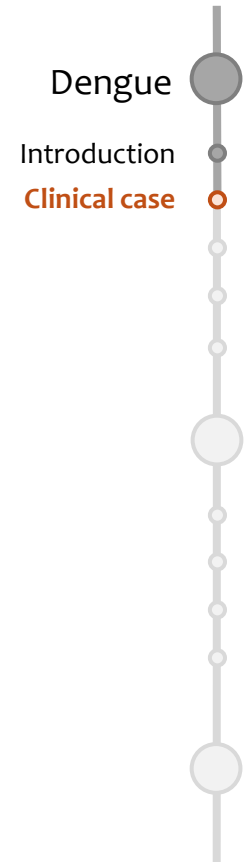
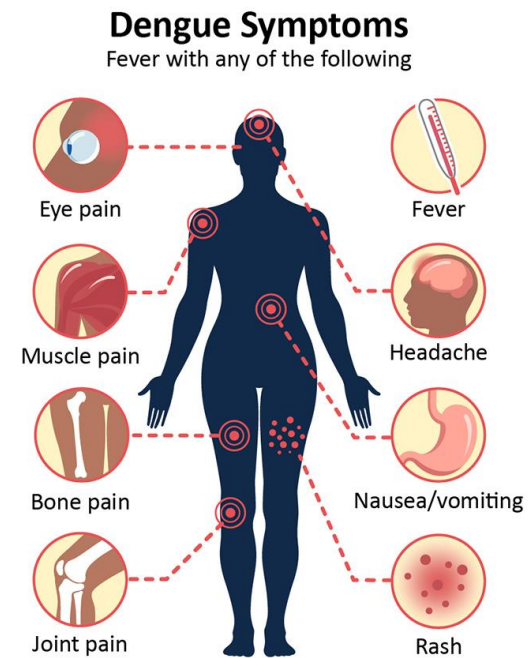
G. Guzman and E. Harris – Dengue – The Lancet (2014)



Problem definition

Clinical case

- Wide spectrum and **non-specific clinical symptoms**; potential confusion with other febrile illness.
- Severe cases may develop **bleeding** and **shock** with potentially fatal outcomes.
- Worse outcomes in **children** and young adults.
- Lack of specific antiviral treatment or effective vaccines; management focuses on **supportive care**.



Problem definition

The dataset used in the study consists of an **aggregation** of prospective clinical data conducted at the Hospital of Tropical Diseases (HTD) and collaborator hospitals in Ho Chi Minh City, **Vietnam** by Oxford University Clinical Research Unit (OUCRU) between **2000 and 2021**.

Code	Year	Population	Type of care	# patients
o6DX	2009-2011	A&C	Inpatient	318
13DX	2010-2014	Children	Outpatient	8107
32DX	2013-2016	A&C	Inpatient and ICU	75
42DX	2016-2018	A&C	Inpatient and ICU	664
DF	1999-2009	Children	Inpatient and PICU	1719
DR	2005-2008	Children	Outpatient	1165
FL	2006-2009	A&C	Inpatient	740
MD	2001-2009	Children	Inpatient	3044
o1NVA	2020-2021	Children	Inpatient	150*

¹ Only children (under 18 years old) have been considered since they were the most represented in the datasets and there are separate paediatric and adult dengue guidelines.

² Dengue diagnosis was defined as one of i) a positive NS1 point of care assay or NS1 ELISA, ii) positive reverse transcriptase-polymerase chain reaction (RT-PCR), iii) positive dengue IgM through acute serology, iv) or seroconversion of paired IgM samples where available.

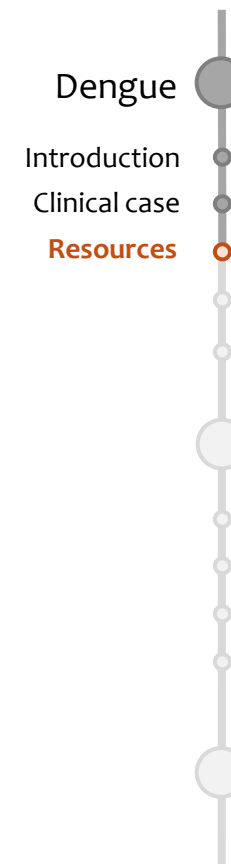
12,884
patients¹

19,516
complete daily profiles

4,344 (33.7%)
diagnosed with dengue²

9
datasets

20
years



Problem definition

Diagnosis

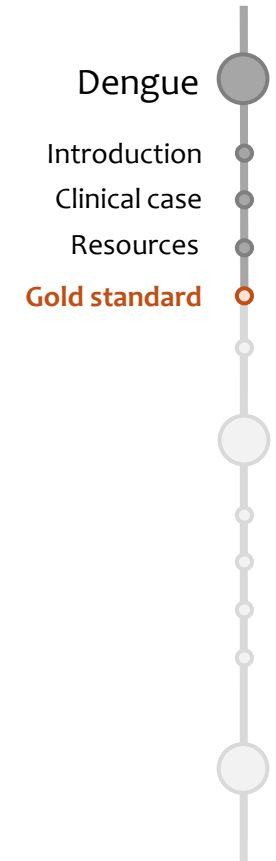
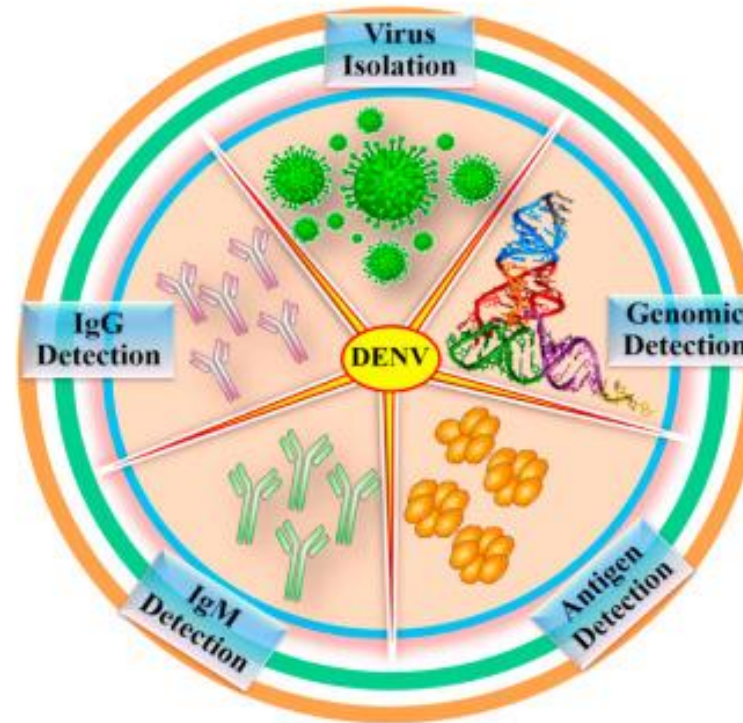
Polymerase Chain Reaction (PCR)
 NS1 Antigen Test
 igM positive through acute serology
 Seroconversion (either single or paired igM and igG)

Classification

WHO 2009
 Probable Dengue: Mild
 Warning signs: Medium
 Severe dengue: Severe

WHO 1997
 Dengue Fever (DF): Standard
 Dengue Haemorrhagic Fever (DHF): Medium
 Dengue Shock Syndrome (DSS): Severe

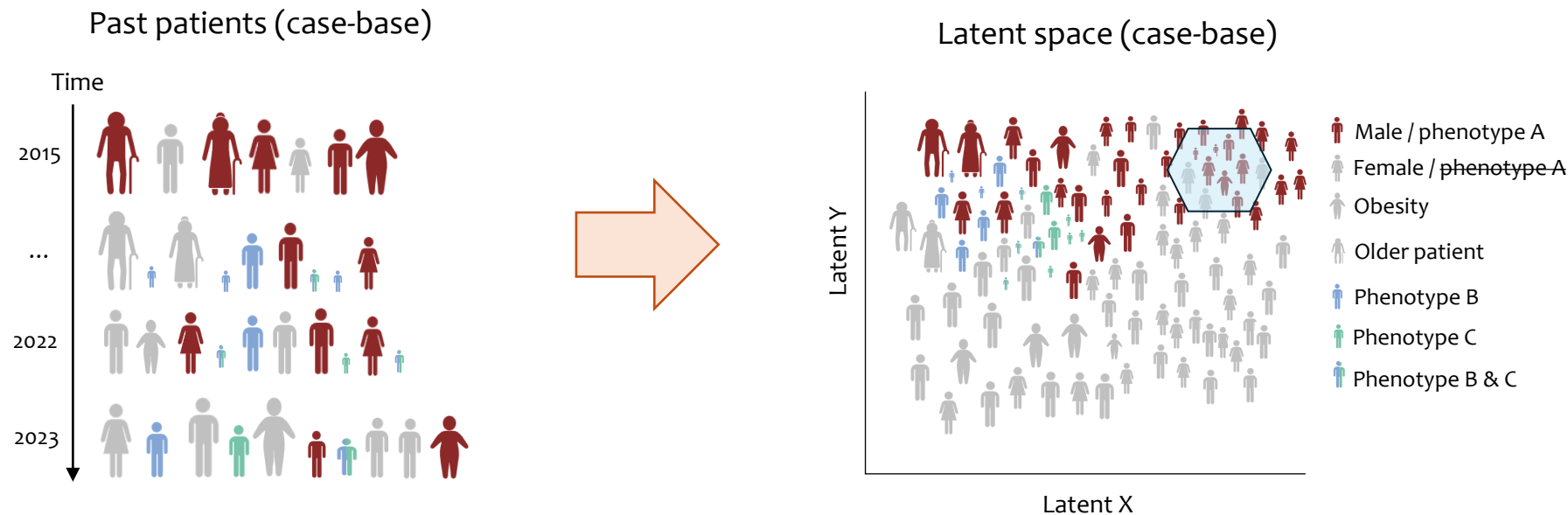
Shifting diagnostic and classification criteria limit supervised approaches



1. S. Subraya et al– *Dengue detection: Advances and challenges in diagnostic technology*. – Biosensors and Bioelectronics (2022)
2. World Health Organization – *Dengue: guidelines for diagnosis, treatment, prevention and control*. – WHO Bulletin (2009)
3. Repository with vital-oucru-clinical documentation: <https://bahp.github.io/vital-oucru-clinical/>

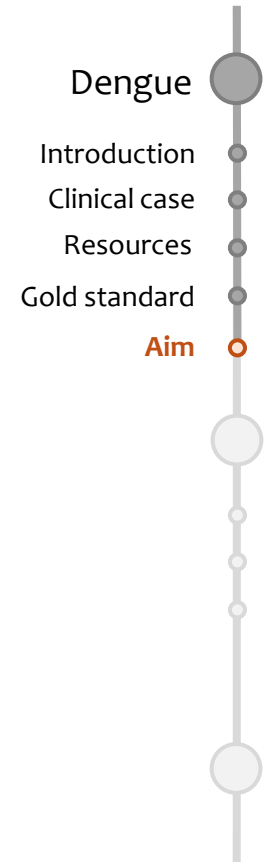
Problem definition

AIM: Development of a **clinical decision support** tool to support clinical management of patients with (or under suspicion of) dengue using **unsupervised techniques** to reduce data complexity and facilitate **visualisation**.



REQUIREMENTS

- Use an unsupervised approach; **no labels**.
- Ability to provide support **from admission**.
- Ability to recalculate and **update on every new input**.
- Especial focus on **visualisation**.



Experimental setup

Choosing your **data wisely** is crucial for model **performance** and feasibility of both **implementation** and **adoption**.

Clinical relevance

Data availability

Quality of data

Frequency of data updates

Cost of data collection

Resource requirements

Turnaround time

Robustness to missing data

Preprocessing requirements

Clinical workflow integration

Interoperability with existing systems

Ethical and Legal considerations

Setting (e.g., LMIC, ICU, ...)

Correlation between variables

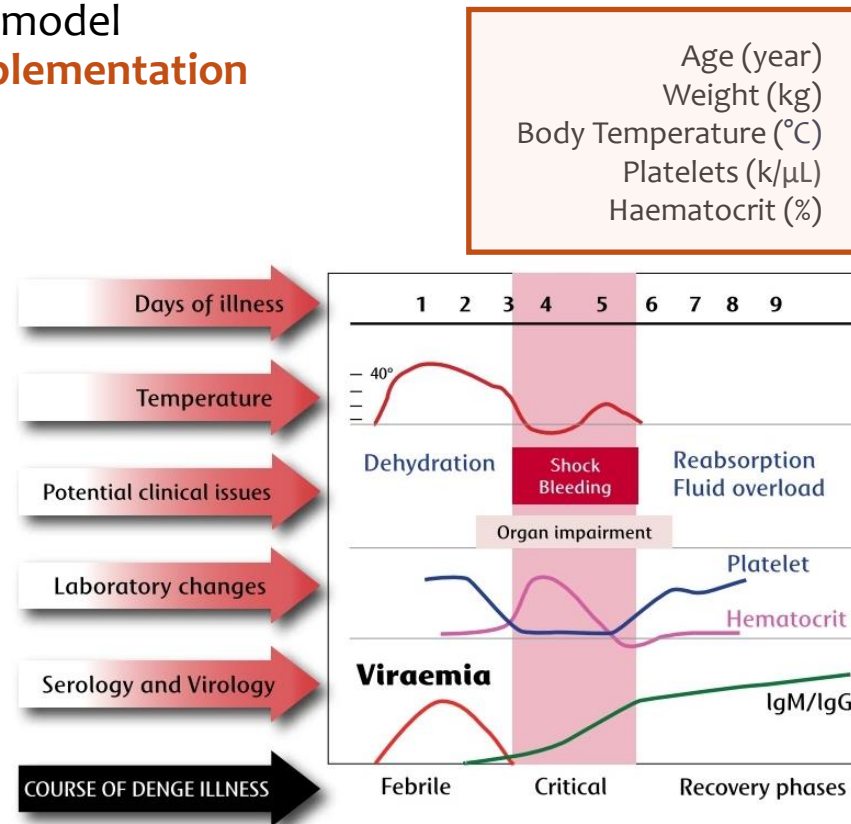


Figure 1. The course of dengue illness diagram. The figure, which has been adapted from WCL Yip, et al 1980 [28], presents phases, lab results, and associated problems.



Experimental setup

Choosing the appropriate **machine learning (ML) approach** according to your data and objectives is crucial for **performance**, **implementation**, and **adoption**. Higher complexity does not necessarily mean better results/adoption.

Labels

Problem type

Data type

Complexity

Interpretability

Resources

Scalability

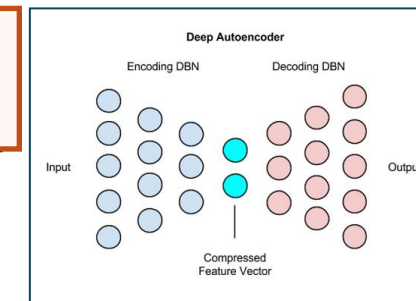
Adaptation

Adoption

Algorithm	Type	Continuous space	Unseen data	Comments
PCA	Parametric	✓	✓	The performance is likely to decrease as dimensionality increases due to its linear nature
T-SNE	Non-parametric	✓		Its inability to be applied to unseen data points makes a real-time similarity retrieval system impossible.
UMAP	Non-parametric	✓		Performance is considerably worse than other approaches for dimensionality reduction.
SOM	Non-parametric		✓	The limitations imposed by the discrete space limits the applicability to similarity retrieval.
Autoencoder	Parametric	✓	✓	The ability to encode unseen samples and its support for higher-dimensional data, including time series and images, make it flexible and idea for similarity retrieval.

PCA: Principal Component Analysis; **t-SNE:** t-distributed Stochastic Neighbour Embedding;
UMAP: Uniform Manifold Approximation and Projection; **SOM:** Self-Organising Maps;

- Complex patterns (non-linear)
- Multiple data sources
- Meaningful representations



Dengue

Introduction

Clinical case

Resources

Gold standard

Aim

Experiment

Feature selection

Algorithm selection

Experimental setup

The **importance** of evaluation **metrics** can fluctuate, often involving a **trade-off**. Therefore, it is crucial to clearly define your **objectives in clinical practice**.

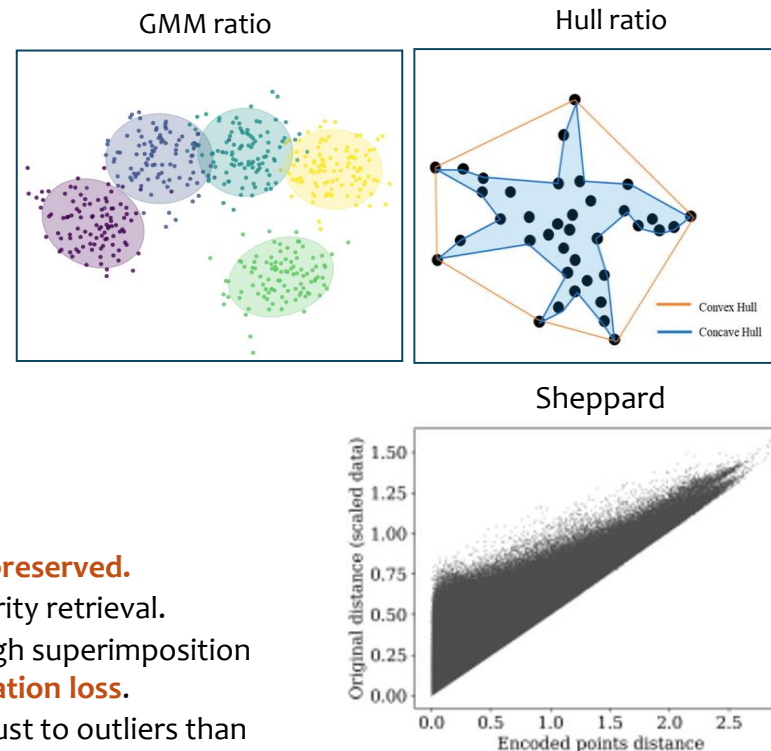
Table 3
Evaluation metrics.

Type	Metric	Aim	
Distance	Sheppard	distance preservation	na
	Pearson	distance preservation	↑
	Spearman	similarity retrieval	↑
	Procrustes	information loss	↓
Density	convex hull ratio	good visualisation	↓
	concave hull ratio	good visualisation	↓
	GMM ratio	good visualisation	↓

GMM: Gaussian Mixture Model

↑ Higher values are better

↓ Lower values are better



Sheppard: Scatter plot of pairwise distances.

Pearson: Linear correlation to verify if **distances are preserved**.

Spearman: Dependence between rankings, for similarity retrieval.

Procrustes: Compare the shape of two objects through superimposition from original to reduced spaces and measure **information loss**.

GMM: Overlapping area within classes. It is more robust to outliers than the convex/concave hull ratio.

Dengue

Introduction

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Aim

Experiment

Feature selection

Algorithm selection

Metrics

Assessment and evaluation

Table 6

Evaluation metrics for various representative hyperparameter configurations

Layers	Activation	Pearson $\uparrow$	Spearman $\uparrow$	Procrustes $\downarrow$	GMM $\downarrow$	Comments
-	-	0.916	0.896	0.272	0.814	PCA
[]	ReLU	0.940	0.920	0.226	0.695	The approximate linearity of the ReLU activation function of this model favours distance preservation.
[]	Sigmoid	0.917	0.906	0.240	0.543	The non-linearity of the Sigmoid activation affects distance metrics slightly and improves density metrics.
[3]	Sigmoid	0.840	0.830	0.301	0.321	It balances distance preservation and density metric results.
[5,4,3]	ReLU	0.635	0.622	0.505	0.104	It is a complex model with good density metric results but produces dense points in the latent dimension not apt for visualisation of patient trajectories over time. In addition, distance metric results show that distances are not preserved and therefore it is inadequate for similarity-based retrieval.

Performance

Outcomes

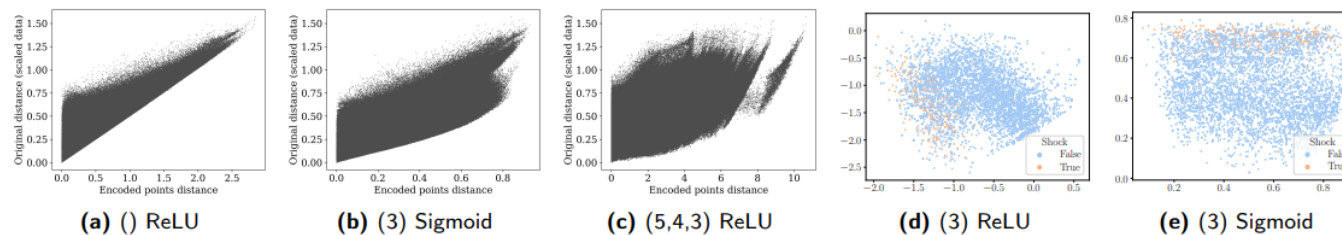
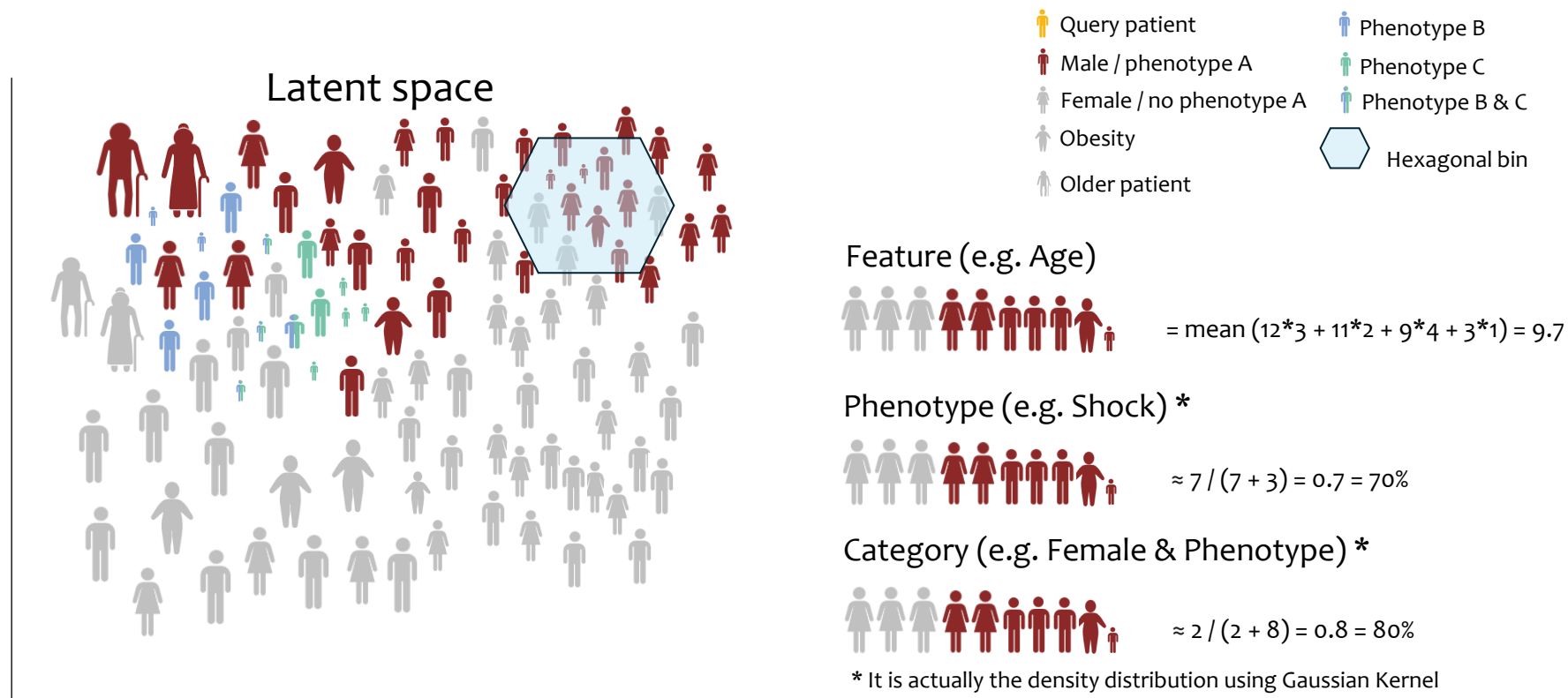


Figure 3: Sheppard diagrams (left) and shock label projections (right). On the left, Sheppard diagrams obtained for autoencoders with three different configurations. On the right, distribution of patients in the latent space with (orange) and without (blue) shock.

Assessment and evaluation

Describing your **algorithm thoroughly** will facilitate **understanding** and might increase **adoption**. Don't just present standard general performance metrics.



Assessment and evaluation

Each point on the graph represents a complete patient's hospital stay, with those experiencing **shock** displayed in orange. They all seem to be grouped on the **top area**.

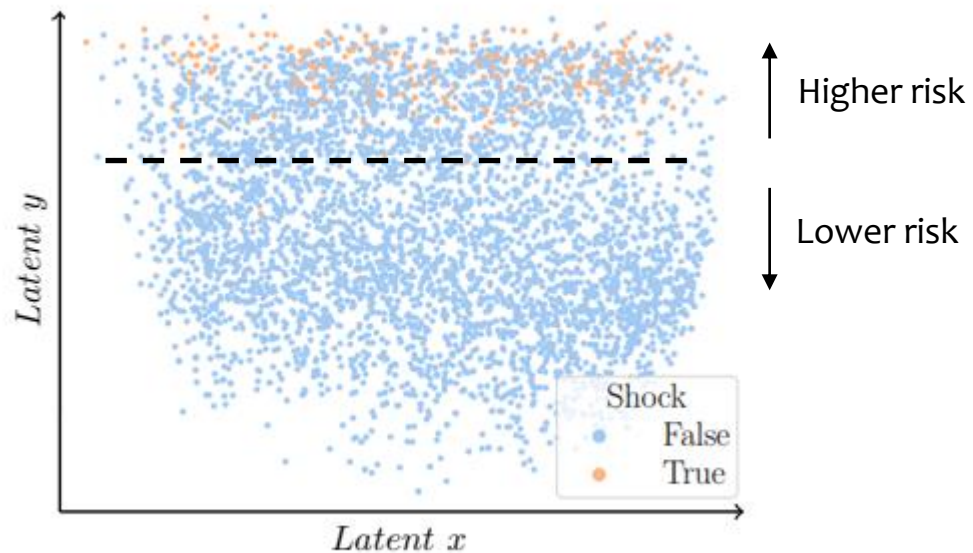
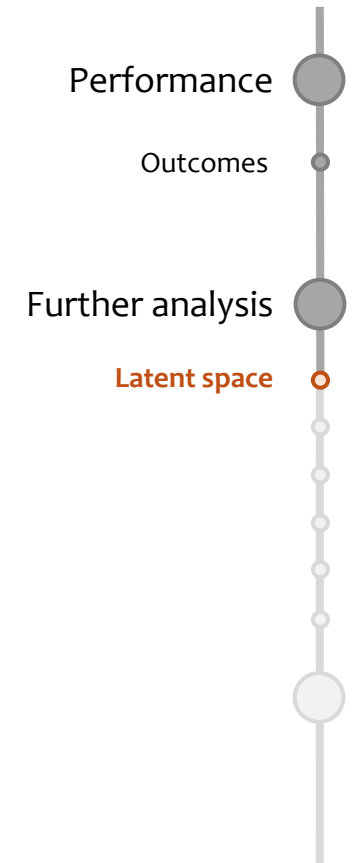


Figure 2. Latent space: embeddings. The worst patient status for all the patients has been projected into the latent space with the shock phenotype.



Assessment and evaluation

Aligns with established characteristics of disease progression such as increase in **haematocrit levels** ↑, decrease in platelet levels ↓ and decrease in body temperature from febrile to critical phase.

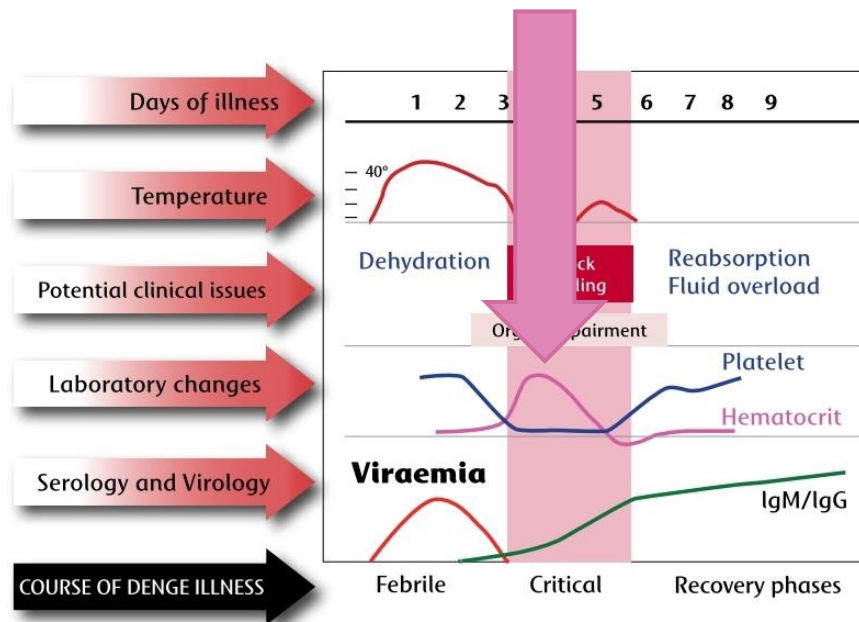


Figure 1. The course of dengue illness diagram. The figure, which has been adapted from WCL Yip, et al 1980 [28], presents phases, lab results, and associated problems.

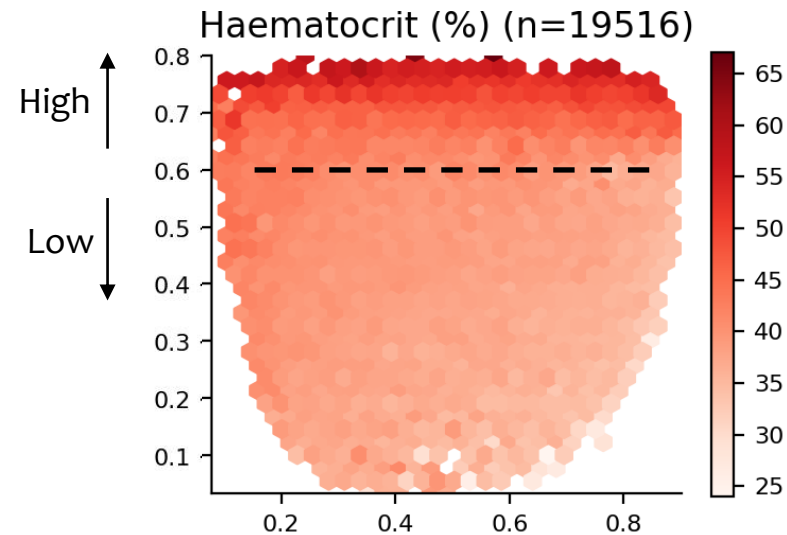
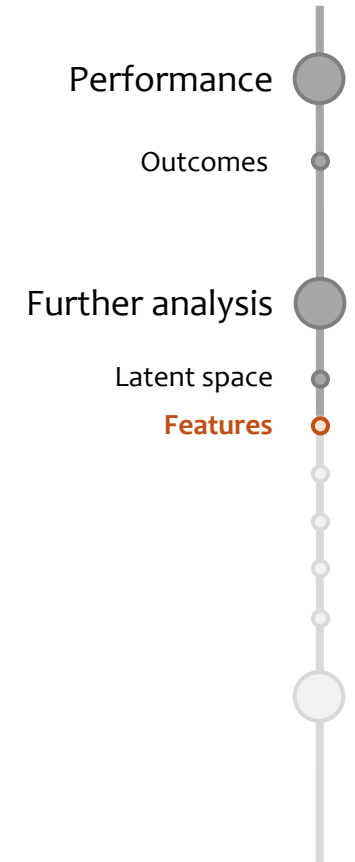


Figure 3. Latent space description: Features. The graphs represent the density distribution using hexagonal binning over the latent space.



Assessment and evaluation

Aligns with established characteristics of disease progression such as increase in haematocrit levels $\uparrow$, **decrease in platelet levels $\downarrow$** and decrease in body temperature from febrile to critical phase.

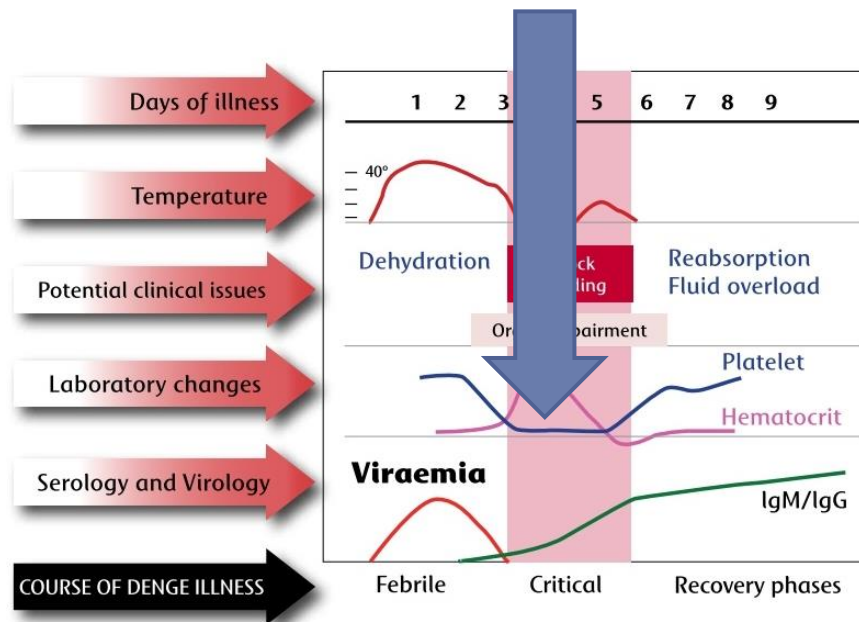


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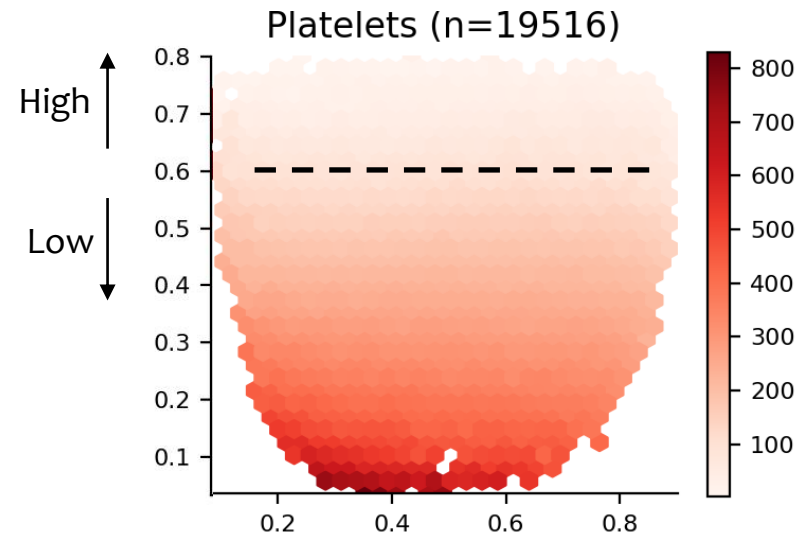
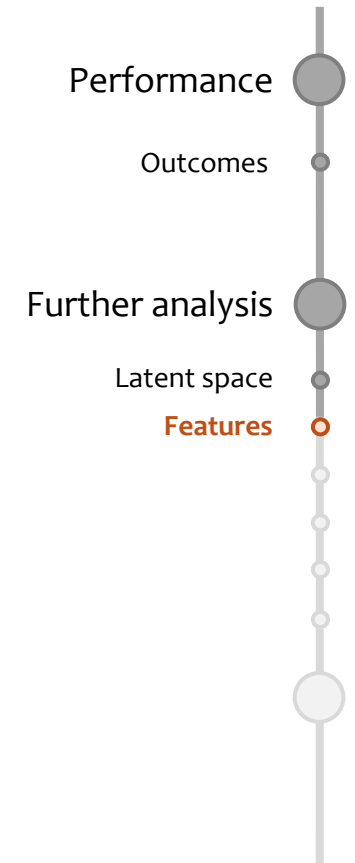


Figure 3. Latent space description: Features. The graphs represent the density distribution using hexagonal binning over the latent space.



Assessment and evaluation

Aligns with established characteristics of disease progression such as increase in haematocrit levels $\uparrow$, decrease in platelet levels $\downarrow$ and **decrease** in body temperature from **febrile to critical** phase.

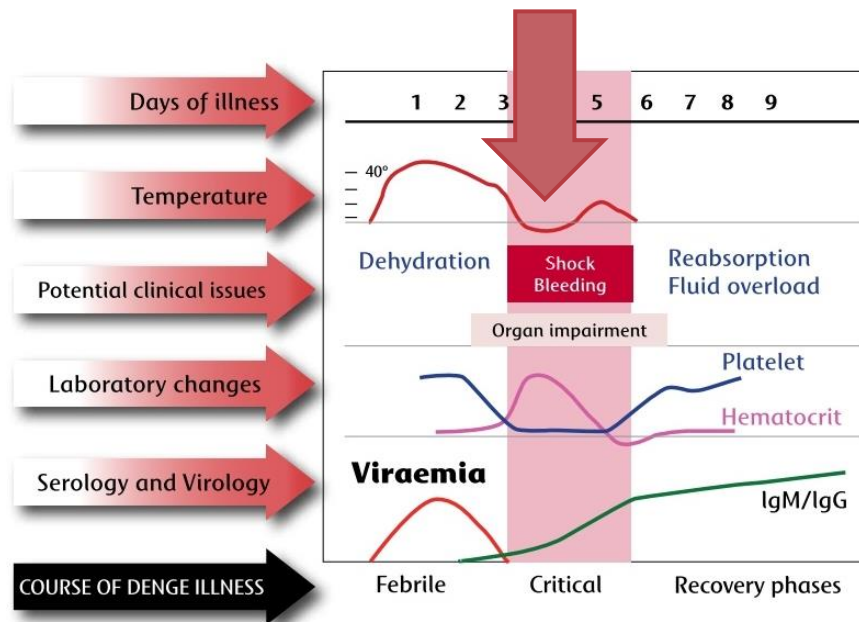


Figure 1. The course of dengue illness diagram. The figure, which has been adapted from WCL Yip, et al 1980 [28], presents phases, lab results, and associated problems.

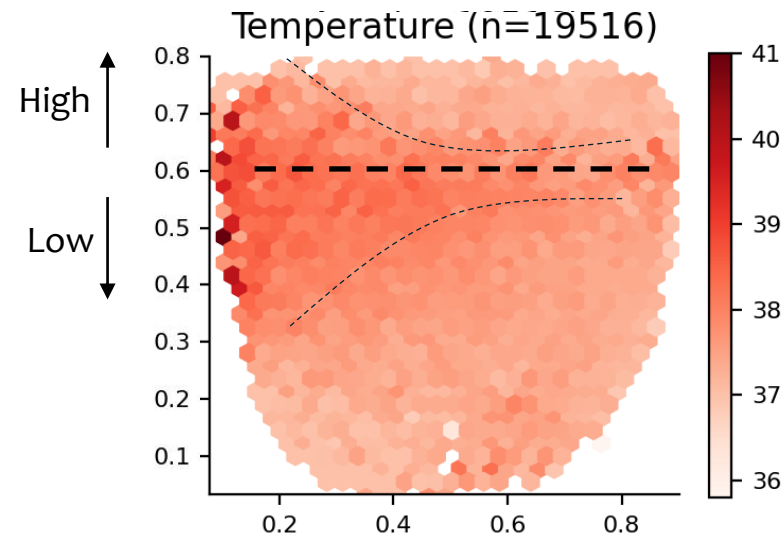
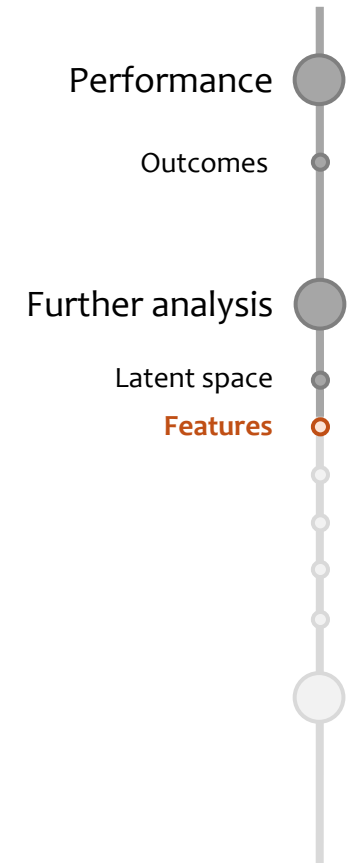
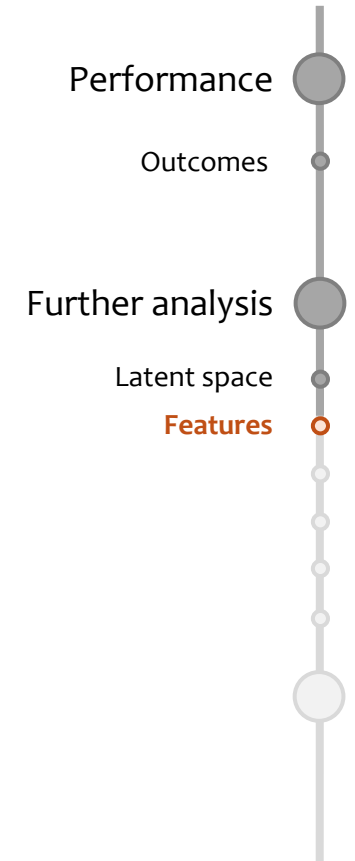
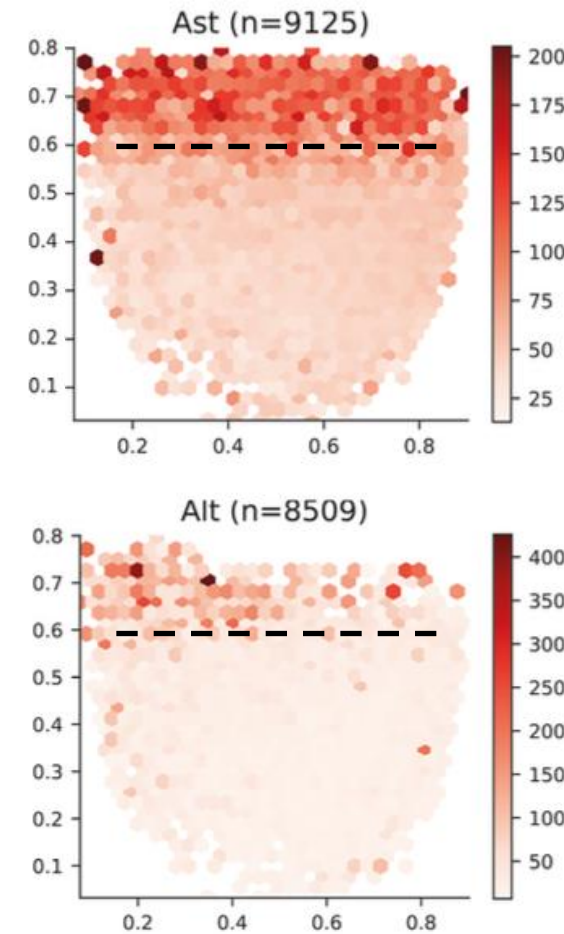
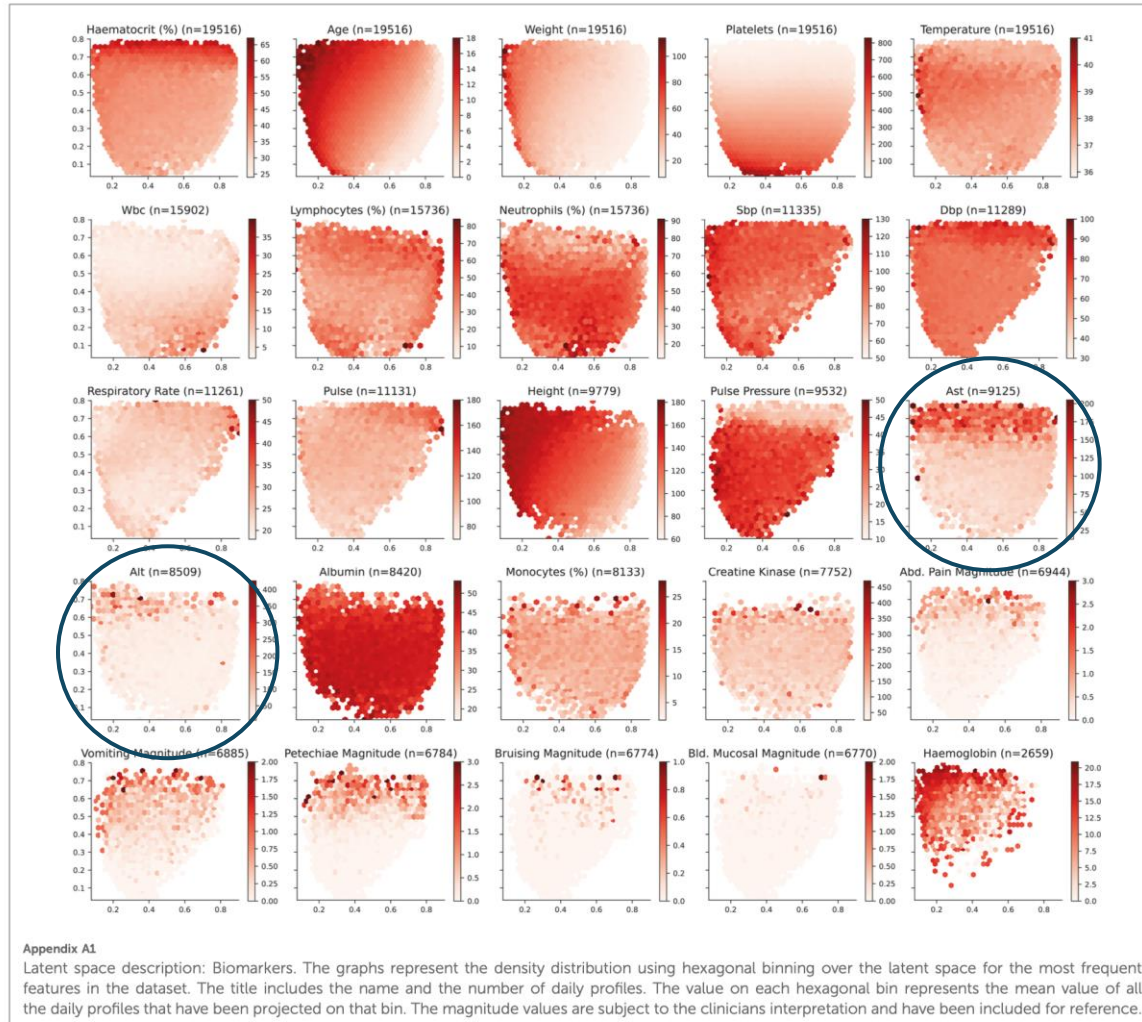


Figure 3. Latent space description: Features. The graphs represent the density distribution using hexagonal binning over the latent space.



Assessment and evaluation



Assessment and evaluation

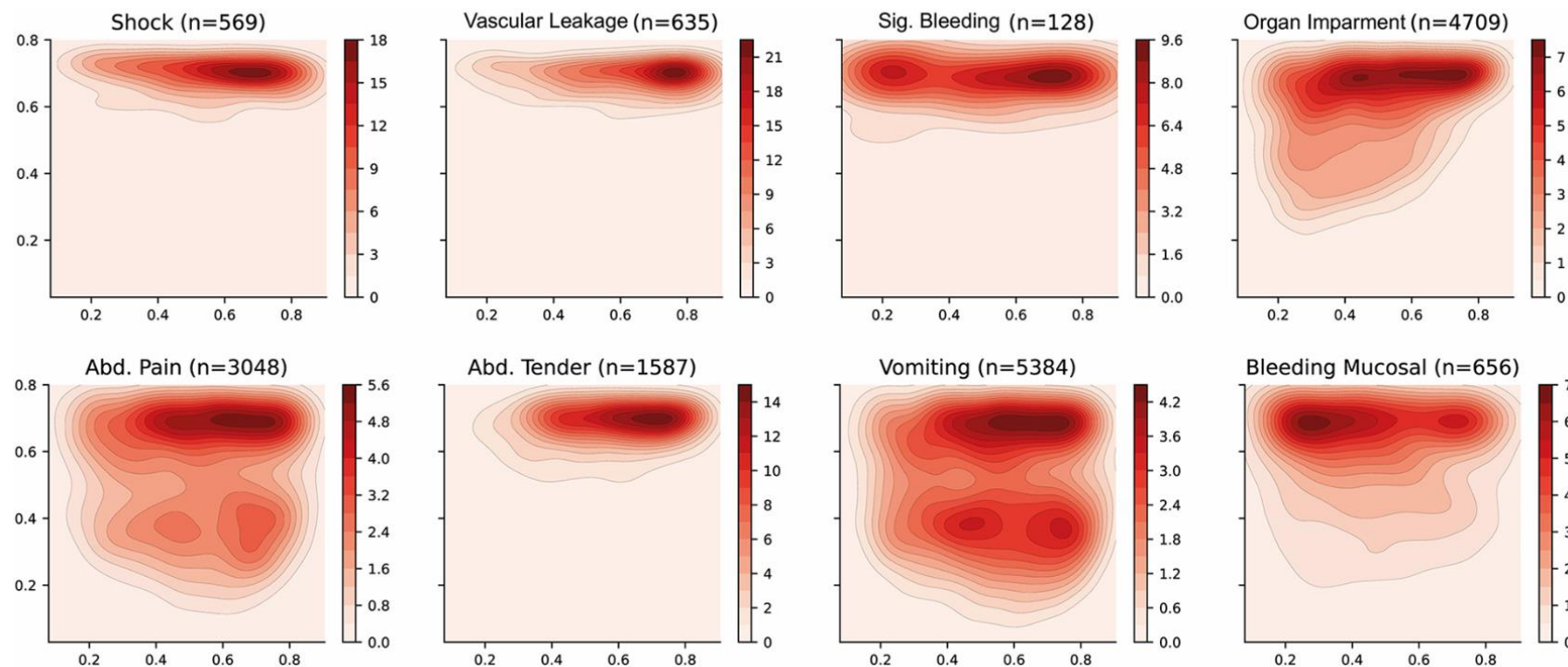
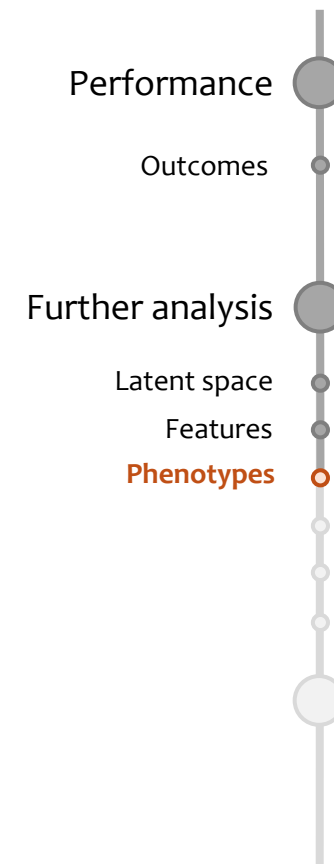


FIGURE 5

Latent space description: Phenotypes. The graphs represent the density distribution using contour lines estimated using a Gaussian kernel over the latent space. The title includes the phenotype and the number of patients in which it occurs. The definition of the compound categories (vascular leakage, significant bleeding and organ impairment) are defined in [Table 2](#). A detailed set of graphs describing other phenotypes has been included in [Figure A2 \(Appendix\)](#).



Assessment and evaluation

Severe dengue

Severe plasma leakage...
 - leading to shock (DSS)
 - fluid accumulation + distress
 Severe bleeding
 Severe organ involvement
 - Liver: AST or ALT ≥ 100
 - Impaired consciousness (CNS)
 - Heart and other organs

Warning signs

- Abdominal pain or tenderness
 - Persistent vomiting
 - Fluid accumulation
 - Mucosal bleeding
 - Lethargy, restlessness
 - Liver enlargement $> 2\text{cm}$
 - Haematocrit $\uparrow$ & Platelets $\downarrow$

Probable dengue

Live in/travel to endemic area
 Fever and 2 of:
 - Nausea and/or vomiting
 - Rash
 - Aches and pains
 - Tourniquet test positive
 - Leucopenia
 - Any warning sign

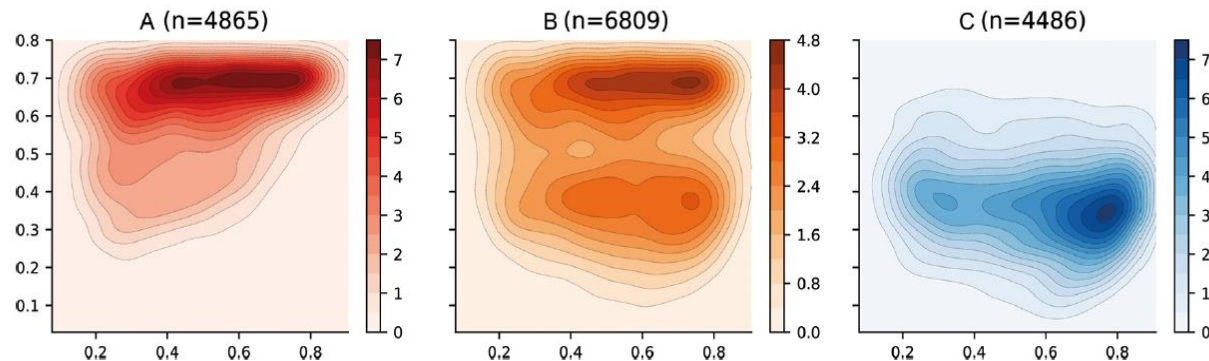
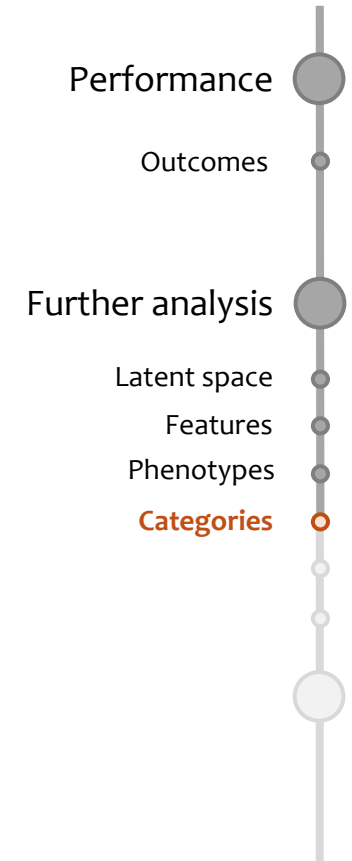
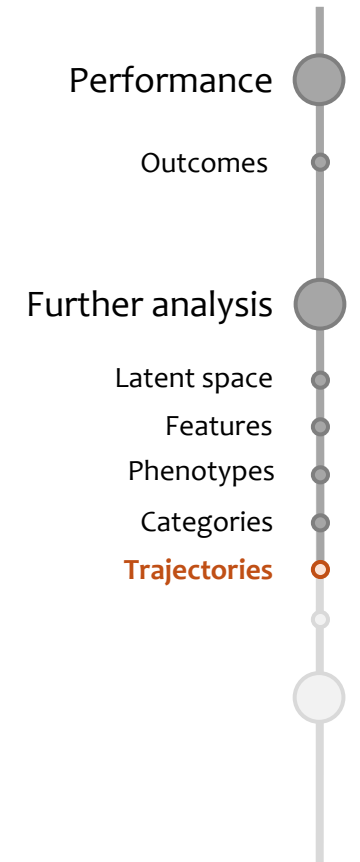
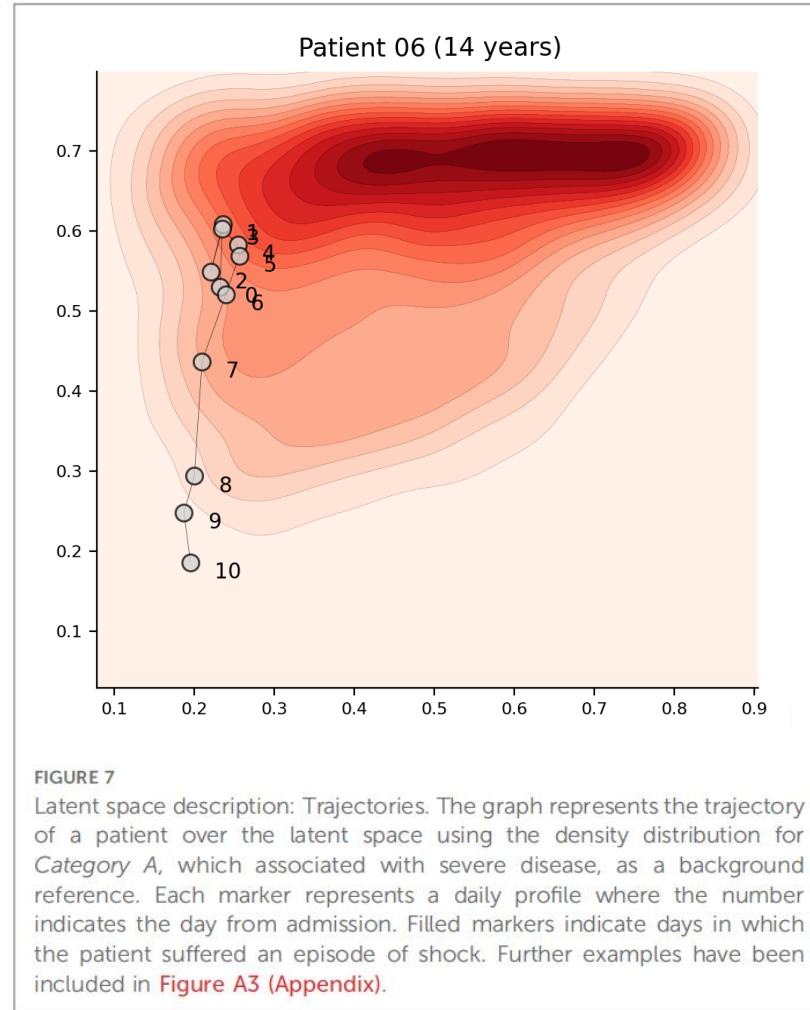
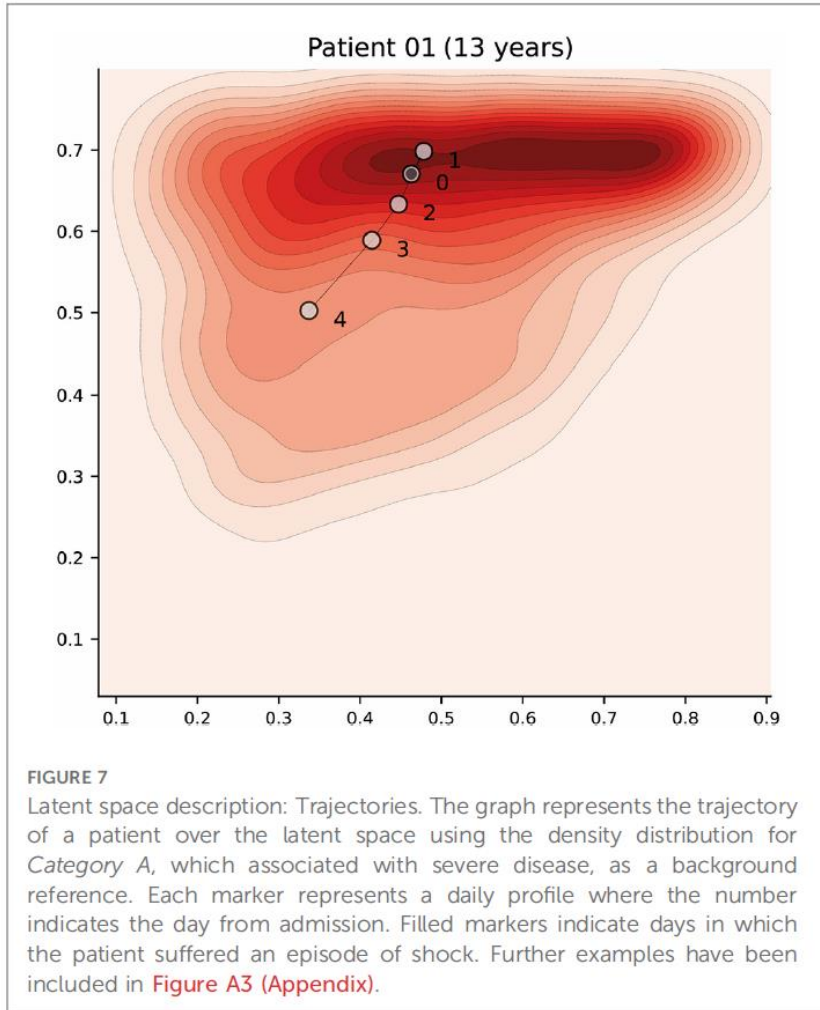


FIGURE 6

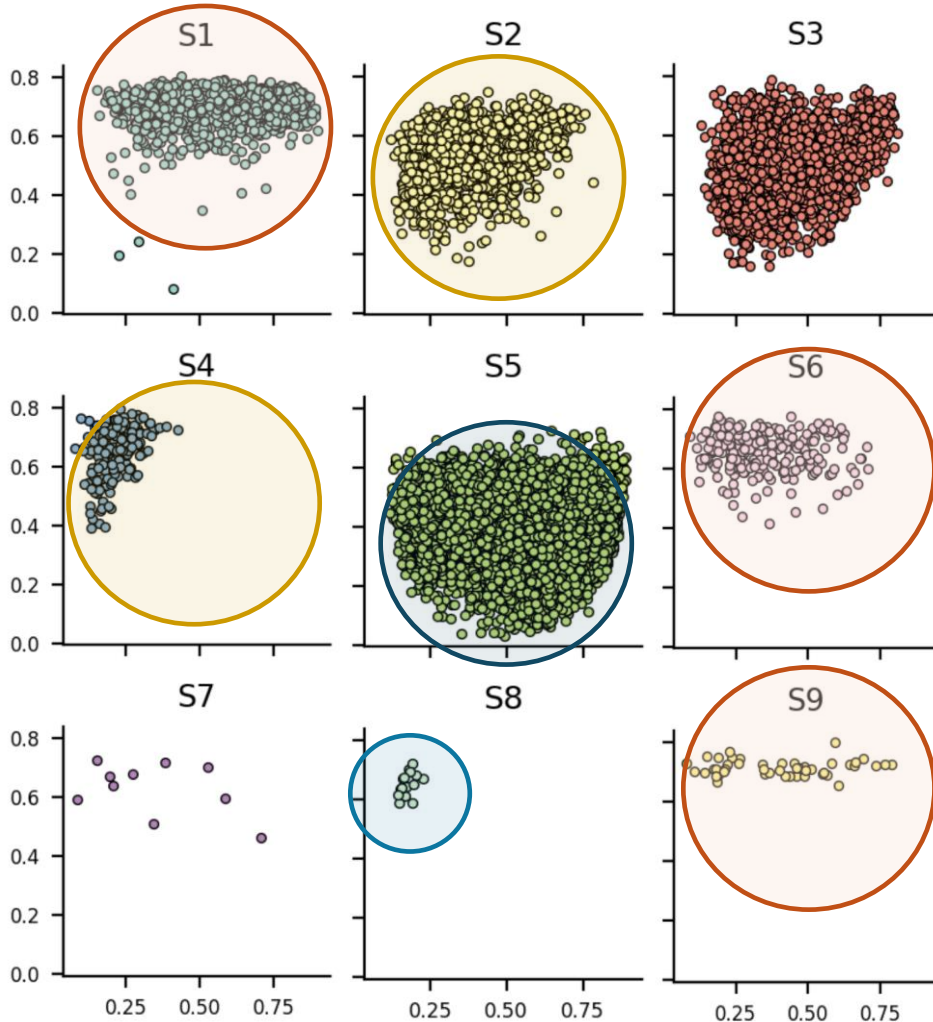
Latent space description: Categories. The graphs represent the density distribution over the latent space for three categories (A, B and C) estimated using a Gaussian kernel. These categories are defined as a compendium of various phenotypes (see Table 2) and therefore the value on each bin (pixel) described in the colorbar represents the estimated density on that bin for which one or more of the conditions associated to the category occurs.



Assessment and evaluation



Assessment and evaluation



Description of studies

S1: Inpatient-based prospective observational descriptive study of clinical features of **DSS** in children and comparison of different fluid solutions for initial resuscitation.

S2: Inpatient-based prospective observational descriptive study examining prognostic factors during the **febrile phase** of non-severe dengue in children.

S3: Outpatient-based prospective observational descriptive study examining prognostic factors during the **febrile phase** of non-severe dengue in children.

S4: An inpatient-based prospective descriptive study of patients with dengue.

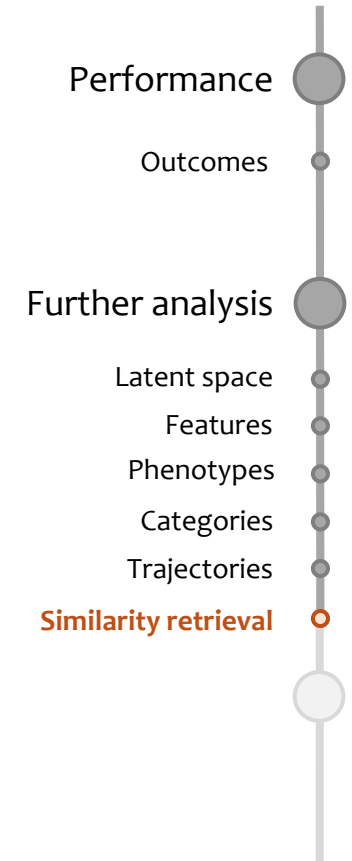
S5: A pilot study to investigate the effects of short course oral corticosteroid therapy in **early dengue** infection in Vietnamese patients.

S6: Laboratory diagnosis and prognosis of **severe dengue**.

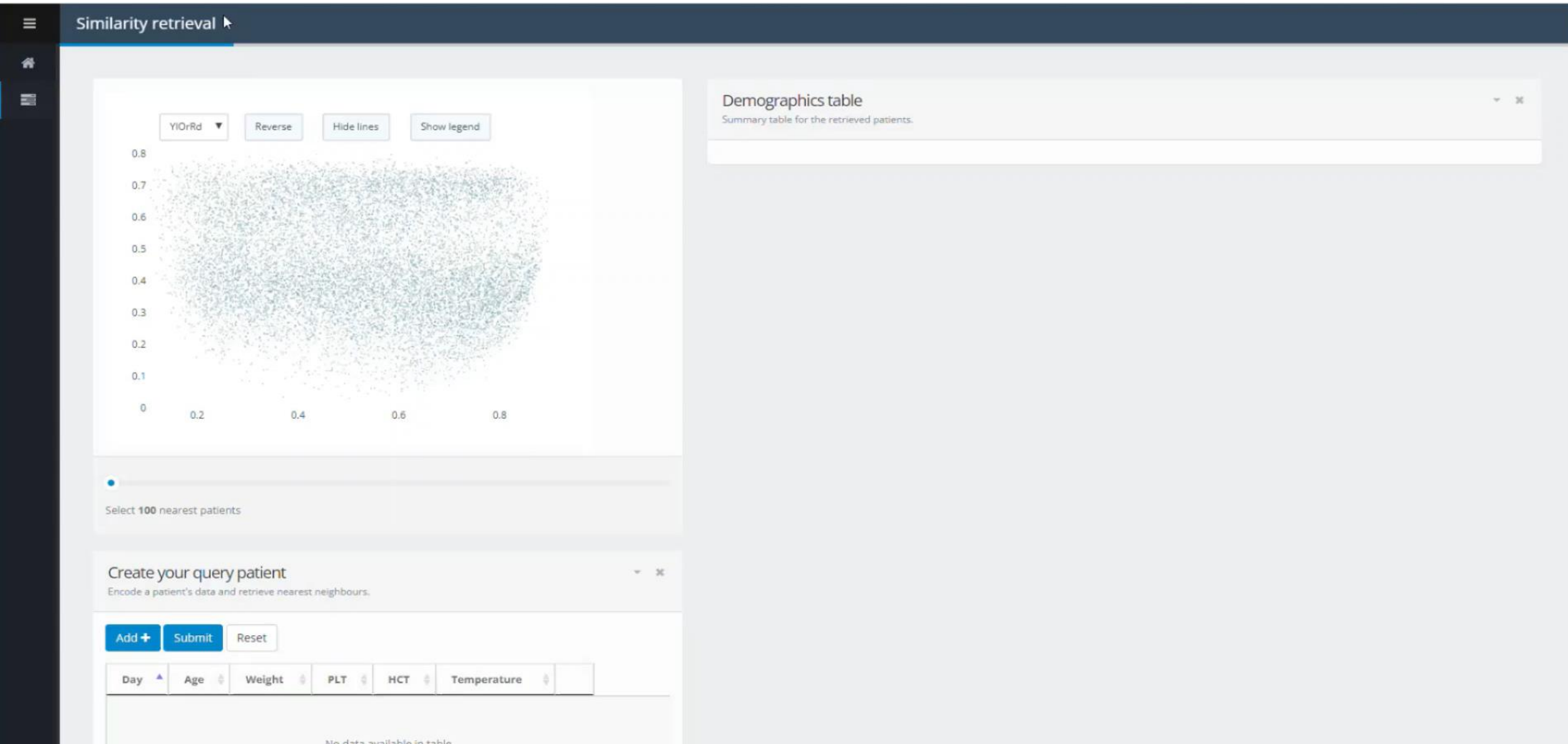
S7: Inpatient-based prospective observational descriptive study to characterise microvascular and endothelial function.

S8: A matched cohort study to characterise the clinical manifestations of dengue in **pregnancy** and investigate the spectrum of adverse maternal and foetal outcomes.

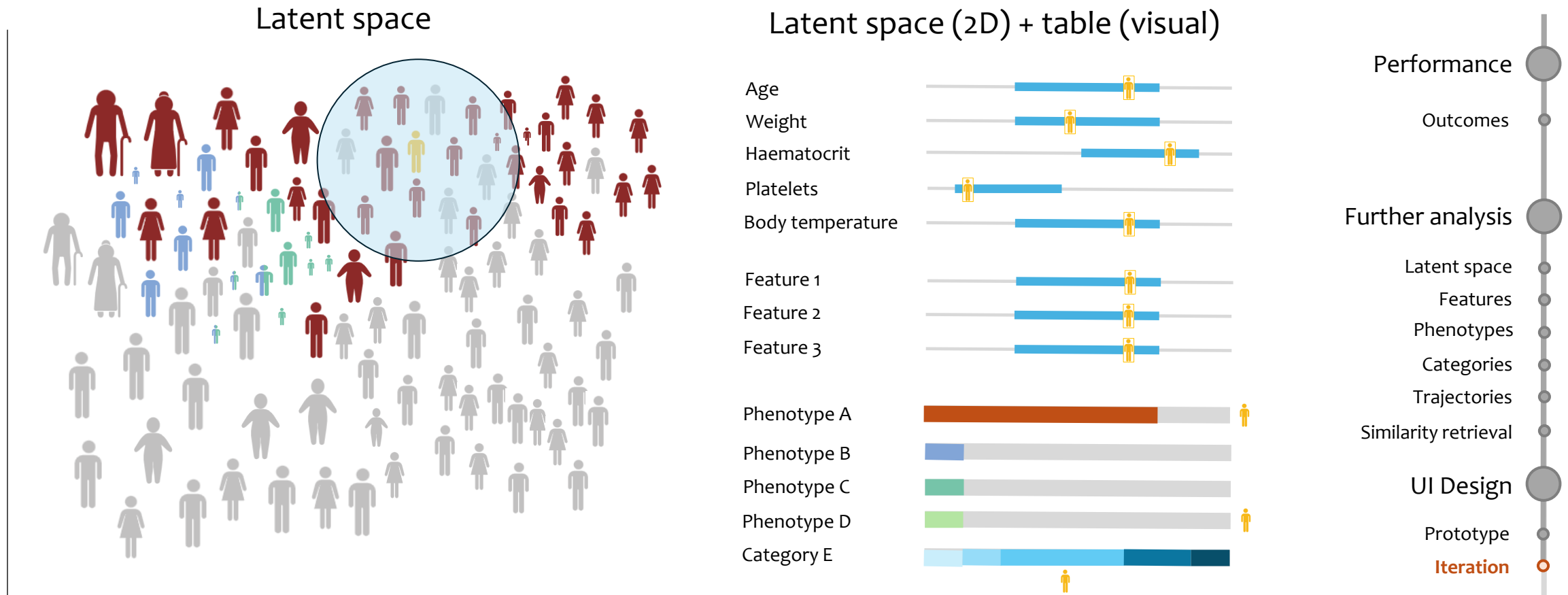
S9: Innovative biomedical engineering and computational science to improve the management of **critical illness** in resource- limited settings.



Initial prototype



UI design for intuitive user experience



The potential secondary use of EHR data

- Descriptive analytics

AMR surveillance from microbiology data

Compute various **resistance metrics** such as resistance rates and trends, multiple resistance, spectrum of activity of antimicrobials, concurrence, or drug resistance rates. These data can **be integrated into machine learning** approaches.



Rates

The proportion of resistant isolates



Trends

Ratio of change of resistance rates over time



Multiple

Ratio of antimicrobials tested to which a pathogen is resistant



Spectrum

#microorganisms that can be treated with an antimicrobial



Concurrence

Whether resistance to one drug is associated to resistance to another



DRI

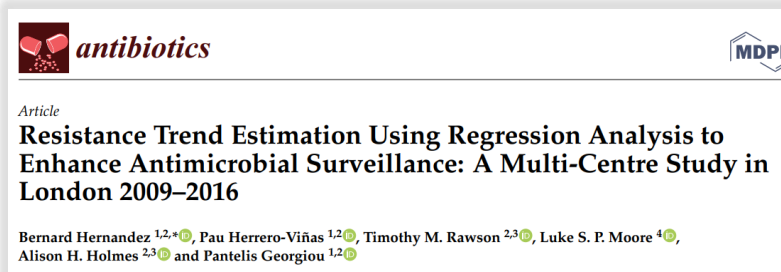
Ratio of resistant isolates to drugs used to treat them



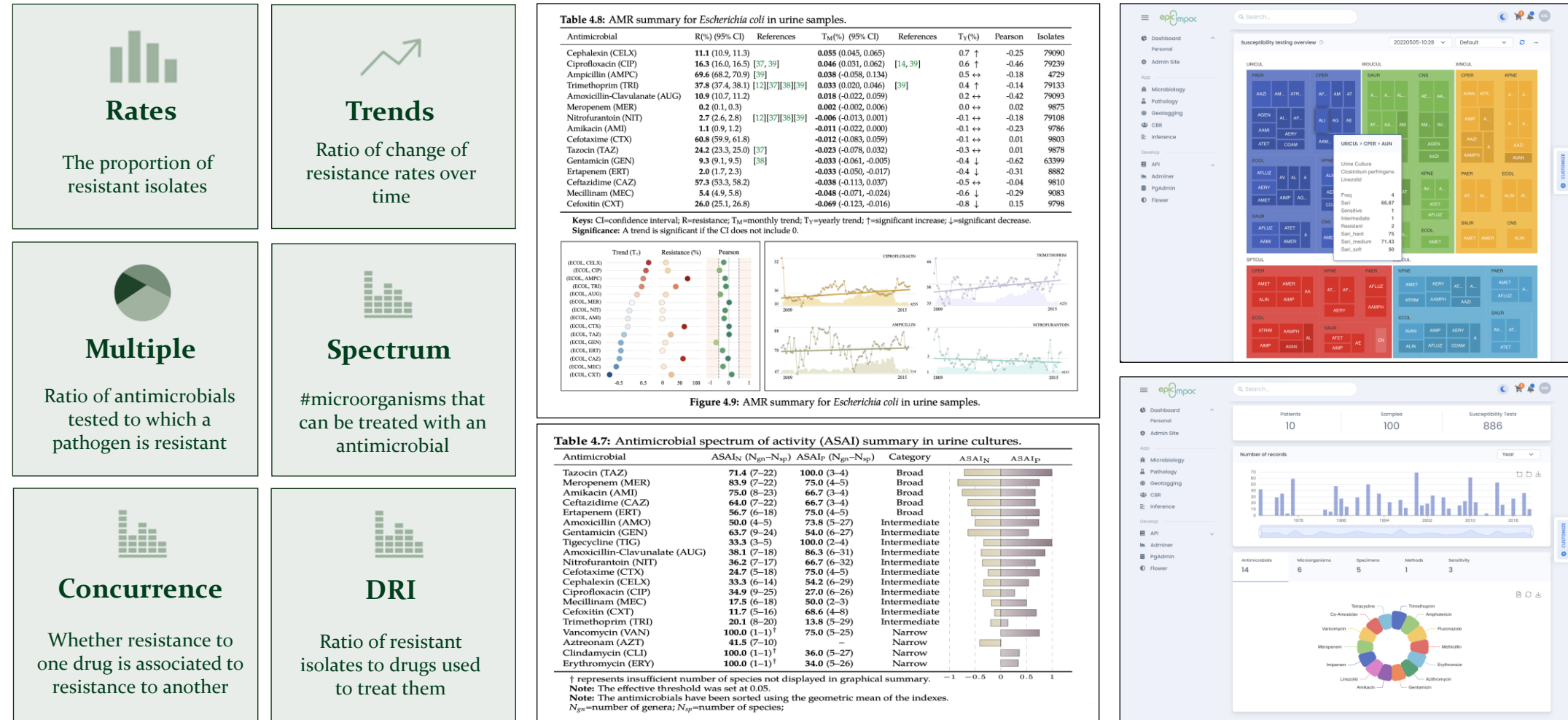
Bernard Hernandez

Research Fellow

Department of Infectious Diseases
Imperial College London



AMR Surveillance



1. B. Hernandez – pyAMR Python library (2023) - <https://bahp.github.io/pyAMR>
2. B. Hernandez –PhD Thesis Imperial College London (2019) – <http://spiral.imperial.ac.uk/handle/10044/1/73000>
3. B. Hernandez et al - *Resistance trend estimation using regression analysis to enhance antimicrobial surveillance: A multi-centre study in London 2009-2016* – Antibiotics (2021)

Unintended consequences of using AI in healthcare

Data bias and discrimination

AI systems can inherit bias present in the training data; thus, **misdiagnoses** or inadequate treatment for underrepresented populations can occur.

Privacy concerns

Collection and analysis of **sensitive patient data**. Risk of unauthorized data breaches, exposing patient information, and violating privacy.

Legal and ethical dilemmas

Determining **accountability** for AI errors and issues when AI-driven decisions conflict with human values and priorities.

Economic disparities

Access to advanced AI may be limited to those who can afford it. This could exacerbate healthcare disparities.

Regulatory challenges

Developing and implementing regulations for AI in healthcare is **complex and slow**. Technological advances may outpace regulatory frameworks.

Validation and accuracy

Ensuring accuracy and reliability of AI algorithms is a challenge. Errors can have serious consequences.

Dependency and skill erosion

Overreliance on AI can **erode the medical skills** of healthcare professionals. In emergency situations or system failures, their ability to provide care may be compromised.

Loss of human touch

Reduction of personal interaction between healthcare providers and patients. Let's remember that these tools are meant to be used by humans!

Conclusions

Enormous potential across the clinical pathway

AI's potential is not limited to one area. It offers **opportunities** to support clinicians at every stage, including diagnosis, prognosis, therapy selection, switch, cessation, de-escalation, monitoring, ...

Data is the foundation and the bottleneck

The value of AI is locked behind data quality. We have a critical need for **standardized**, interoperable data (like EHRs) to build reliable and deploy evidence-based tools in clinical settings.

There is a major need to gather **real-time**, non-invasive, and accurate data in such a way that fits seamlessly into the clinical workflow.

From prediction to action

Prognostic models aren't crystal balls. Their true value is in identifying **high-risk** patients early, enabling clinicians to move from reactive care to proactive intervention

The “co-pilot” model

The goal is **augmentation**, not replacement. We must co-design systems with clinicians and carefully address ethics, bias, and trust to create truly helpful tools.

Thank you!

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<https://bahp.github.io/website/>

7th of November 2025

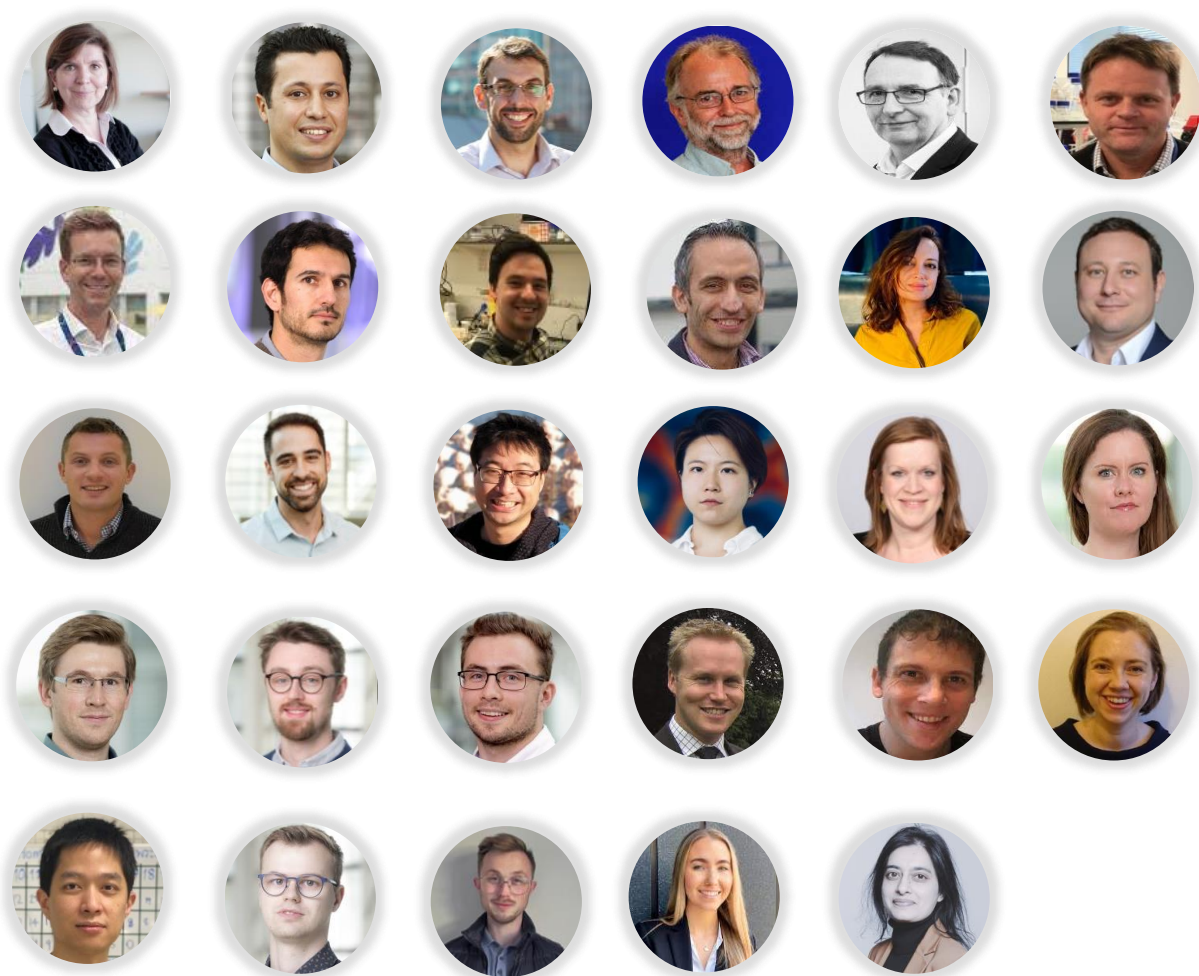


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NHS Trust



The infection management pathway

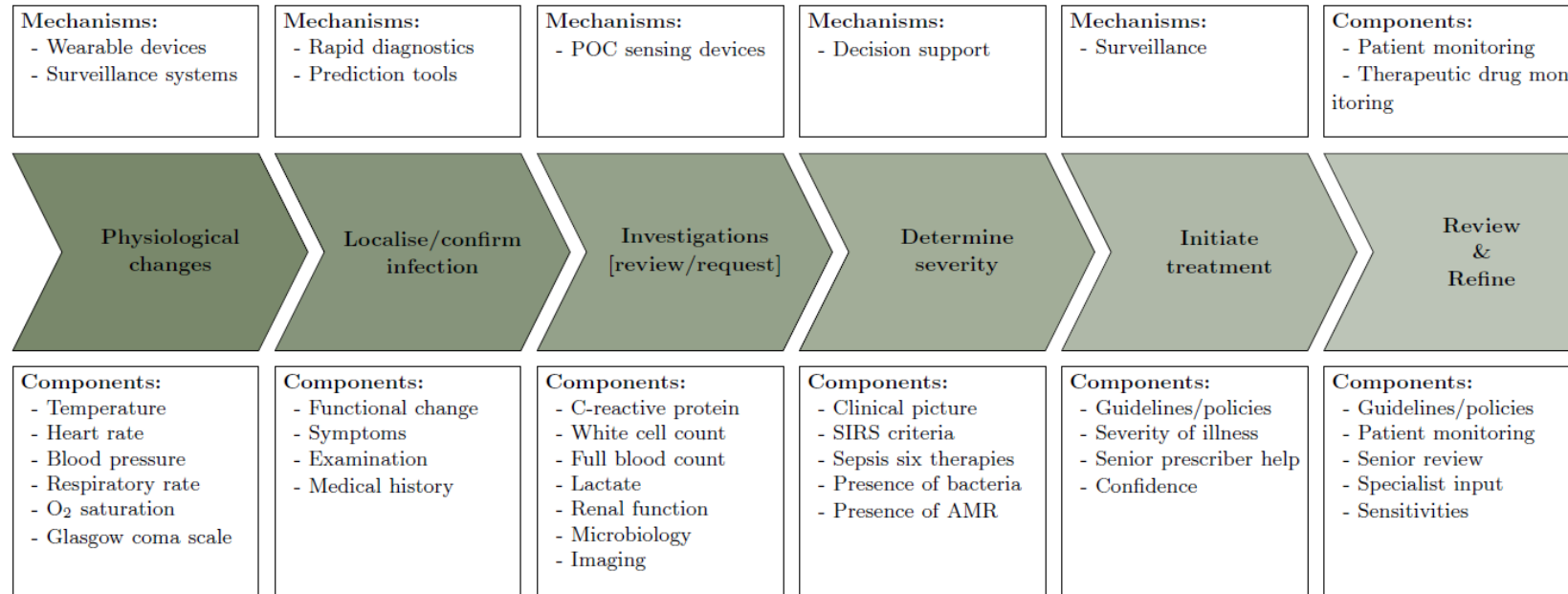


Figure 7.1: Acute infection management pathway. The stepwise Bayesian like infection management pathway followed by clinicians as described in [5]. Boxes below describe parameters commonly required during such phases of the infection management pathway. Boxes above suggest methods, systems or techniques to improve data collection on such phases. The module(s) providing support on each of the steps are: case-based reasoning (CBR), probabilistic inference (PI) and antimicrobial resistance surveillance (AMR).



Supervised learning

6, 5, 4	3	2	1
Infection >500.000 daily profiles 2.7% prevalence 6 biomarkers CRP, WBC, BIL, CRE, ALT, ALP SVM AUC ROC 0.85 (95% CI:0.84 - 0.86) SENS 0.75 SPEC 0.91 104 patients 35% prevalence at 72h 42% had microbiological tests AUC ROC 0.84 (95% CI:0.76 - 0.91) SENS 0.89 SPEC 0.63 at 0.81	SARS-CoV-2 1186 patients 65% prevalence 14% had microbiological tests 21 biomarkers Full Blood Count (FBC) SVM AUC ROC 0.91 (95% CI:0.76 - 0.91) SENS 0.80 SPEC 0.89 54 patients 52% prevalence AUC ROC (0.96% CI: 0.90 - 1.00) SENS 0.75 SPEC 0.90	Dengue 8100 patients 27.7% prevalence Age, Gender, Day + 4 biomarkers HCT, PLT, WBC, LYMPH XGB AUC ROC 0.86 (95% CI:0.84 - 0.86) SENS 0.92 SPEC 0.56 PPV 0.73 NPV 0.84 + seasonality!	Dengue Shock Syndrome 4131 patients 5.4% prevalence Age, Gender, Weight, Day + 2 biomarkers first 48h HCT, PLT ANN AUC ROC 0.83 (95% CI:0.76 - 0.85) SENS 0.66 SPEC 0.84 PPV 0.18 NPV 0.98
Bloodstream Infection using temporal dynamics trough LSTM			

1. Damien et al – *The diagnosis of dengue in patients presenting with acute febrile illness using supervised machine learning and impact of seasonality* – Frontiers in Digital Health (2022)
2. Damien et al – *Applied machine learning for the risk-stratification and clinical decision support of hospitalised patients with dengue in Vietnam* – PLOS Digital Health (2022)
3. TM Rawson et al – *Supervised machine learning to support the diagnosis of bacterial infection in the context of COVID-19* – JAC Antimicrobial Resistance – BMC Medicine (2021)
4. B Hernandez et al – *Data-driven web based intelligent CDSS for infection management at the point of care* – PhD Thesis Imperial College London (2019)
5. TM Rawson et al – *Supervised machine learning for the prediction of infection on admission to hospital: a prospective observational cohort study* – JAC (2018)
6. B Hernandez et al – *Supervised learning for infection risk inference using pathology data* – BMC Medical Informatics and Decision Making (2017)